# Tech pack: CORRIE F rev 3 (door D2) - 5200 cu in hunting pack bag

Revision 3, door D2, crown kit M shown (S / L cut lists in cutting_plan.csv). Issued 20261113. Generated by designs/pack_5200/mfg_corrie_f_rev3.py from the pattern module; patterns, DXF and markers in ../patterns/. Units mm, N, g (imperial alongside).

| Item | Value |
|---|---|
| Pieces | 76 piece types; 77 fabric / sheet pieces; 75 webbing / cord / elastic |
| Ledger reconciliation | 102 rows, 0 unmatched, 0 pieces without a row; hardware rows unmapped: 0 |
| Mass from pattern areas | 567.2 g vs ledger 567.7 g (-0.08 %, limit 3 %); bag ledger base 1132.6 g / as hunted 1414.4 g / meat trip 1610.4 g (team B basis); primary fabrics +94.9 g |
| Thread | 18.28 g in the bag from the op lengths (no cone waste); ledger books 18.27 g; purchase quantities in the BOM include 15% cone waste |
| Operations | 90 ops; beginner 74.6 h; factory SAM 508 min |
| Cost a bag | materials $367 primary / $367 alternate; + CMT: US $907, Asia $444 (ASSUMPTIONS) |
| Load points | 26; lowest SF 1.93 (B-17 IM3 tab perimeter (double row)); every load point is released by a proof test (section 8) |
| HOLD | none |
| Pattern flags | none |
| Volatile (re-run on a pattern change) | pattern: B-22 x; pattern: F-01; pattern: FP width; pattern: hinge window vs B-18; pattern: ledger; pattern: ledger (US-chan) |
| Fabric sets | PRIMARY = today's fabrics (build this). ALTERNATE = team B set (ULTRAGrid body, 70D collar, ULTRA 200X floor) only after gates ALT-1..3 and FS-O5. |
| Owner rules | No hook-and-loop; coil zips; mostly solid colours, camo on accents only; width <= 394 mm (15.5 in); 85 L +/- 3. |

Contents: 1 cover; 2 equipment and stitch standards; 3 flats / piece maps; 4 BOM; 5 cutting plan; 6 operation sequence; 7 load points and sketches; 8 QC, release tests, defects; 9 packaging; 10 costing; 11 delta register; 12 assumptions.

# 2. Equipment and stitch standards

| ID | Machine / tool |
|---|---|
| WF1 | Single-needle compound-feed walking foot lockstitch, Juki DU-1181N (or LU-1508N). Prototype: any walking foot that runs Nm 110 + tex 90 cleanly (e.g. a portable walking-foot machine). |
| WF2 | Heavy single-needle walking foot lockstitch, large hook, Juki DNU-1541S (or LU-1561/2210 class): webbing stacks, Hypalon, tex 135 box-X. Prototype: WF1 hand-cranked through the thick spots. |
| BT | Computer bartacker Juki LK-1900BN (ISO 301 path / 304 zigzag, programmable, 40 x 30 mm field): every 25 mm box-X and the slit bartacks. Prototype: box-X on WF2 with the J1 template. |
| PS | Programmable pattern sewer Juki AMS-210EN (220 x 100 mm field) with a webbing clamp: the 38 mm box-X that do not fit the LK-1900 field (HE, T3, B-24). Prototype / small shop: WF2 with the J1 template. |
| ZZ | Zigzag lockstitch (ISO 304), Juki LZ-2290 class or a portable straight / zigzag walking foot: elastic laps. |
| LASER | CO2 laser cutter 60-100 W, bed >= 600 x 900 mm, fume extraction: Hypalon, TPU-840D, B-12, port bindings, HDPE <= 0.8 mm (or shear + round the corners). |
| HK | Hot knife, blade type, on a steel / glass plate with the J8 length stop: webbing, elastic, cord. |
| SK | Straight knife, 5 in blade (Eastman Blue Streak II class) on a cutting table for production lays. Prototype: 45 / 60 mm rotary cutter, 600 x 900 mat, 6 x 24 in ruler. |
| DIE | Clicker press + steel-rule dies: only above about 500 bags a year (Hypalon tabs, B-12, keepers). |
| TAPE | Hot-air seam-tape machine: NOT used. Rev 3 has no taped seams (ASSUMPTION NO-TAPE: the bag is water resistant, not waterproof; contents go in the user's dry bag). Needed only for a waterproof variant. |

| Use | Thread (record) | Needle | SPI (mm) | Stitch |
|---|---|---|---|---|
| 40D-100D, mesh, silpoly | bonded_nylon_tex70 | Nm 90/14 DPx17, round point R / Nm 90/14 DPx17, medium ball SUK (knit mesh) | 10 (2.5) | 301 |
| body seams, zips, Hypalon perimeters, B-11 sew-down | bonded_nylon_tex90 | Nm 110/18 DPx17, round point R | 8 (3.2) | 301 |
| load box-X | bonded_nylon_tex135 | Nm 140/22 DPx17, round point R | 7 (3.6) | 301 path (LK-1900 / AMS-210EN programmed) |
| bartacks, elastic laps | bonded_nylon_tex90 | Nm 110/18 DPx17, round point R | LK-1900 std | 304 |

Round point (R) or slim set point on every coated fabric, laminate, Hypalon and webbing. Never a leather / cutting point (LR, S, DI): it cuts the reinforcement and the coating and drops seam strength. Ball point (SUK) only on the power mesh. New needle every bag (prototype) or every 8 h (production).

Seam efficiency: a 301 seam is taken at 80% of the weaker strip (clothsim/fabrics.yaml, attach_system.SEAM_EFF); box-X capacity = 0.5 x 1.5 x thread break / stitch length x path (attach_system.box_x, ETA_BOX 0.5 ASSUMPTION).

| Code | Seam construction |
|---|---|
| SSa2 | SSa plain seam, 12 mm allowance, then both allowances turned to one side and caught by a 2nd row at 6 mm (ISO 4916 1.01.01, two rows; no binding, see ASSUMPTION NO-BIND) |
| SSa | SSa plain seam, single row (ISO 4916 1.01.01) |
| SSa-ins | SSa with a webbing tab / loop inserted between the plies (ISO 4916 class 1 with insert); tab ends reach the cut edge, a backtack each side of the tab |
| LSd | LSd lapped patch: piece laid flat on the panel, perimeter row(s) (ISO 4916 class 2, lapped) |
| EFa | EFa edge turned once and edge-stitched (ISO 4916 class 6, edge finish) |
| EFb | EFb edge turned twice, stitched (ISO 4916 class 6); hem round a stiffener strip |
| ZIP | zip set: each tape laid face down on its panel edge, sewn at 5 mm, turned and edge-stitched at 3 mm (ISO 4916 class 1, two rows per tape) |
| BOX | box-X: rectangle + both diagonals through the webbing stack (ISO 4915 301 path; ISO 4916 class 7 when it fixes the webbing to a panel) |
| BT | bartack, ISO 4915 304 zigzag lockstitch, LK-1900 standard pattern |
| DART | dart: fold on the dart centre, sew leg to leg, tie off at the apex (ISO 4916 1.01.01) |

| Jig | What |
|---|---|
| J1 | Box-X templates, clear acrylic 2 mm, one per box size in the load table (section 7 / SK-1), slots for chalk on the rectangle and both diagonals; production: LK-1900 / AMS-210EN programs of the same paths |

| Jig | What |
|---|---|
| J2 | Swell window corner template + clip board (8 swells, notch to notch) |
| J3 | B-27 port gauge: 6 mm turn, 20 mm lap, facing line; a slotted ruler |
| J4 | Webbing fold gauge: aluminium angle with stops at 15 / 30 / 45 / 50 / 60 / 100 mm |
| J5 | LK-1900 / AMS-210EN work clamps for 25 and 38 mm webbing stacks (Hypalon + panel) |
| J6 | Door zip board: B-02door + B-02u laid flat with the corner-tangent notches pinned to the board |
| J7 | Lid dart template (B-29a, 4 darts, apex drill marks) |
| J8 | Hot-knife length fence with a scale in mm and in (cut lengths in cutting_plan.csv) |

# 3. Flats and piece maps (from the pattern pieces)

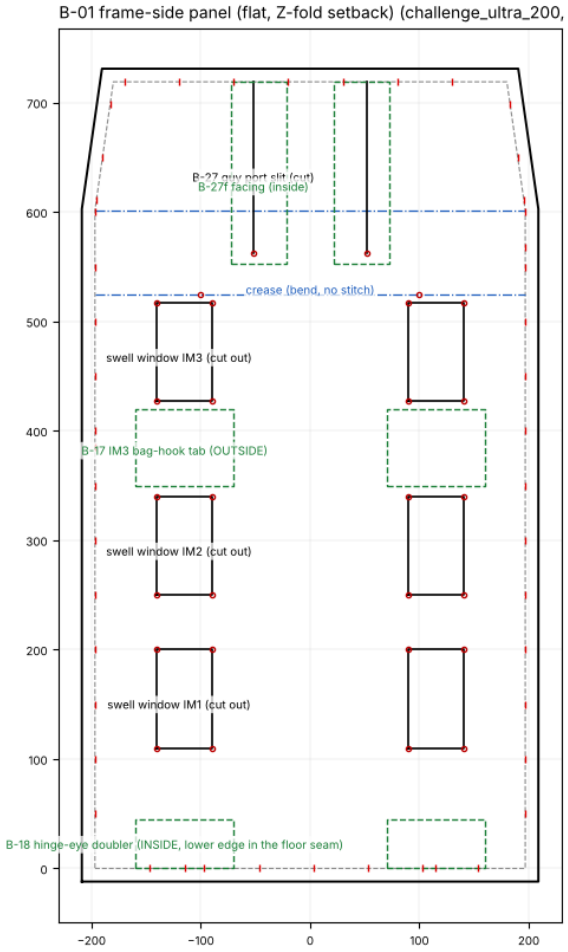


**Side profile, 6-day load (45 L)**

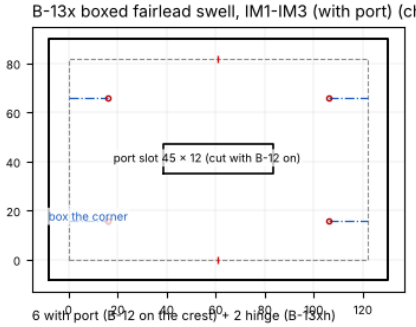
CORRIE F rev 3 (D2), H1 frame, belt B. Crown straps over the lid. Rendered 3D figure, not a photograph

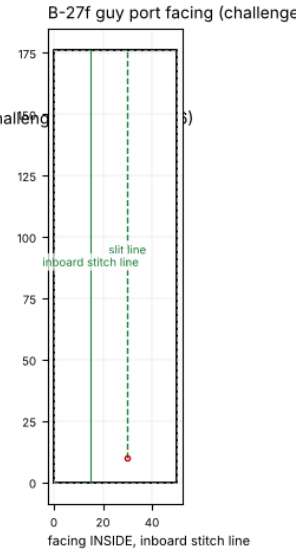
Render of rev 3 (docs/screens/20261112_BL030_2.png, BL-030): side view, lid and crown straps. Colours / camo placement as drawn there.

SK-3  B-01 frame-side panel: flat work in order 4.02 swells, 4.03 B-27 ports, 4.04 B-17 (outside), 4.05 B-18 + HE (inside), 4.06 I2K



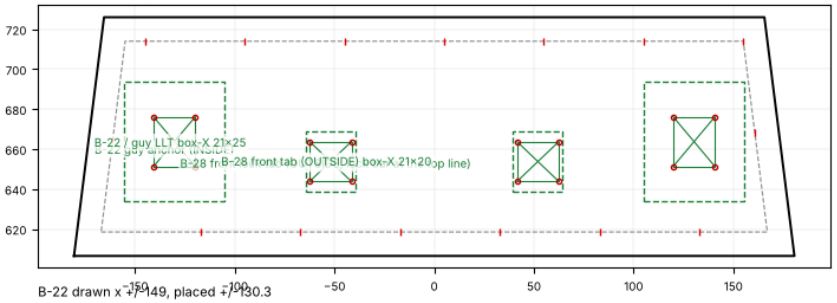

6 swell windows (cut-outs), B-27 slits x +/-52 z 595-745. Volatile: hinge window vs B-18 (read from the pattern).



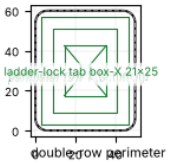
SK-4  Hinge band B-02h: B-22 INSIDE at x +/-130.3 (volatile), LLT box-X, B-28 front tab box-X (outside); then the door hinge seam

B-02h D2 hinge band (z 650 to the collar seam) (cordura_330d_nylon66, cut 1)

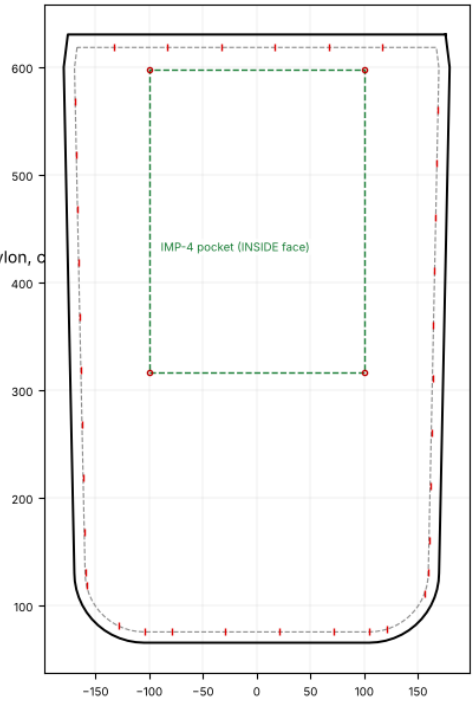



B-22 drawn x +/-149, placed +/-130.3

B-22 guy corner tab (hypalon_coated_nylon, c



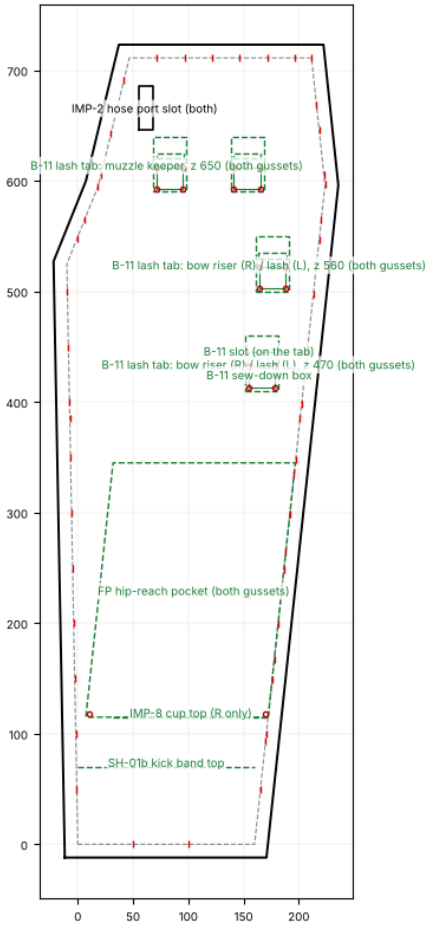

B-02door D2 door (whole aft face z 110-650, R 55 corners) (cordura_330d_nylon




IMP-4 placement INSIDE

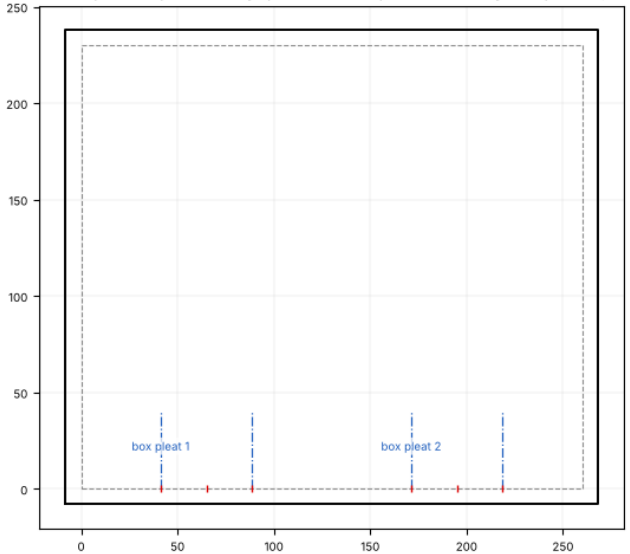
SK-5  Side gusset B-03 (cut L+R): kick band, FP pocket, IMP-8 (R only), hose slot, 4 B-11 lash tabs (op 4.07), K-W keeper notches on the aft edge; corner seams carry HPL (L) and E5 (R)

B-03 side gusset (cordura_330d_nylon66, cut 2)



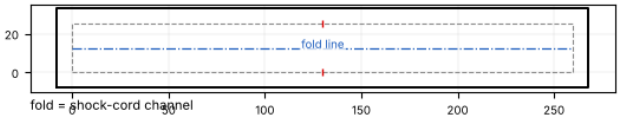

FP z 150-380

FP-a hip-reach pocket body (power mesh) (power_mesh_nylon_spandex, cut 2)




flat 260 wide, 2 box pleats

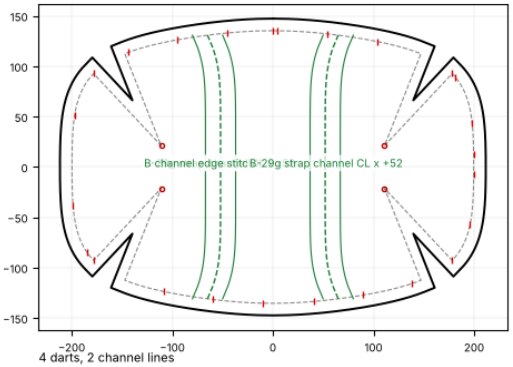
FP-b hip-reach pocket mouth band (camo) (polyester_ht_630d_printable, cut 2



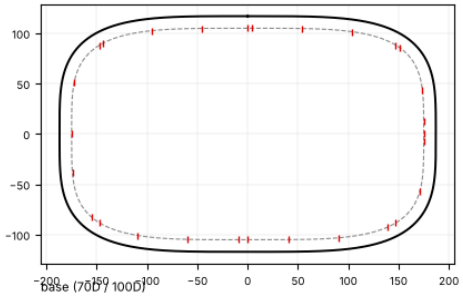

fold = shock-cord channel

SK-6  B-29 crown lid: darts → channels → zip between the aft strips → caps → crown|wall ring → wall|base ring (turn through the zip)

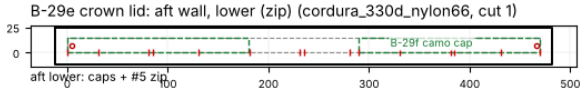
B-29a crown lid: crown (4 corner darts) (cordura_330d_nylon66, cut 1)



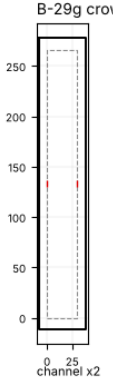

B-29b crown lid: base (lies on the roll) (robic_100d_ripstop, cut 1)




B-29e crown lid: aft wall, lower (zip) (cordura_330d_nylon66, cut 1)




B-29g crown lid: strap channel (co



# 4. Bill of materials (one bag, full build, crown kit M)

Production qty = 10-bag marker + 3 % lay loss where part A nested one, else net / utilisation (ASSUMPTION). Prototype (1-bag marker) quantity in the Proto column. Waste % = gross / net - 1. Thread: purchase quantity includes the cone waste; the grams in the bag (notes) do not.

| Section | Item | Material id | Qty | Unit | Imperial | Waste % | $ / unit | $ | Alternate / kit | Proto qty | Source |
|---|---|---|---|---|---|---|---|---|---|---|---|
| fabric | challenge_ultra_200: B-01x1 B-06fx1 B-13xx6 B-27fx2 | challenge_ultra_200 | 0.487 | m | 0.53 yd | 14.2 | 65.6 | 31.94 | same / base | 0.743 | record challenge_ultra_200; net 0.606 m2 from the pattern areas; 10-bag marker eff 0.92 + 3 % lay loss (part A WASTE_PROD) |
| fabric | cordura_330d_nylon66: B-03x2 B-02doorx1 B-02hx1 B-07x1 IMP8-ax1 F-02x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1 | cordura_330d_nylon66 | 0.824 | m | 0.90 yd | 23.7 | 15.3 | 12.61 | cordura_330d_nylon66 / ultragrid_210d_uhmwpe_ripstop / addon,as_hunted,base | 1.135 | record cordura_330d_nylon66; net 1.015 m2 from the pattern areas; 10-bag marker eff 0.88 + 3 % lay loss (part A WASTE_PROD) |
| fabric | cordura_500d_nylon66: B-05x1 B-16x2 | cordura_500d_nylon66 | 0.092 | m | 0.10 yd | 34.6 | 16.4 | 1.50 | challenge_ultra_200 / base | 0.22 | record cordura_500d_nylon66; net 0.104 m2 from the pattern areas; 10-bag marker eff 0.81 + 3 % lay loss (part A WASTE_PROD) |
| fabric | hdpe_sheet: B-07sx2 MS-sx2 | hdpe_sheet | 0.047 | m2 | 0.51 ft2 | 42.9 | 4.58 | 0.21 | same / base,meat_trip | 0.052 | record hdpe_sheet; net 0.033 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags) |
| fabric | hypalon_coated_nylon: B-17x2 B-18x2 B-22x2 B-11x8 | hypalon_coated_nylon | 0.055 | m2 | 0.59 ft2 | 42.9 | 27.9 | 1.53 | same / base | 0.096 | record hypalon_coated_nylon; net 0.038 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags) |
| fabric | polyester_ht_630d_printable: B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1 | polyester_ht_630d_printable | 0.305 | m | 0.33 yd | 127.6 | 19.7 | 6.00 | same / addon,as_hunted,base | 0.738 | record polyester_ht_630d_printable; net 0.201 m2 from the pattern areas; 10-bag marker eff 0.47 + 3 % lay loss (part A WASTE_PROD) |
| fabric | power_mesh_nylon_spandex: FP-ax2 | power_mesh_nylon_spandex | 0.129 | m | 0.14 yd | 42.9 | 10.9 | 1.41 | same / base | 0.246 | record power_mesh_nylon_spandex; net 0.136 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags) |
| fabric | robic_100d_ripstop: B-06x1 B-09cx1 IMP4-ax1 IMP4-bx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1 | robic_100d_ripstop | 0.958 | m | 1.05 yd | 17.8 | 17.5 | 16.77 | nylon_ripstop_70d / robic_100d_ripstop / addon,as_hunted,base,meat_trip | 1.208 | record robic_100d_ripstop; net 1.239 m2 from the pattern areas; 10-bag marker eff 0.90 + 3 % lay loss (part A WASTE_PROD) |
| fabric | silicone_coated_poly_ripstop_40d: F-01ax1 F-01bx2 MS-ax1 | silicone_coated_poly_ripstop_40d | 1.404 | m | 1.54 yd | 42.9 | 10.9 | 15.36 | same / as_hunted,meat_trip | 1.715 | record silicone_coated_poly_ripstop_40d; net 1.498 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags) |
| fabric | tpu_coated_nylon_840d: B-12x6 B-09px2 IMP8-bx1 | tpu_coated_nylon_840d | 0.036 | m2 | 0.39 ft2 | 42.9 | 16 | 0.58 | same / base | 0.09 | record tpu_coated_nylon_840d; net 0.025 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags) |
| fabric_alternate | challenge_ultra_200: B-01x1 B-05x1 B-06fx1 B-13xx6 B-27fx2 B-16x2 | challenge_ultra_200 | 0.587 | m | 0.64 yd | 17.6 | 65.6 | 38.52 | challenge_ultra_200 / cordura_500d_nylon66 / base | 0.743 | record challenge_ultra_200; net 0.710 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85 |
| fabric_alternate | cordura_330d_nylon66: IMP8-ax1 F-02x1 | cordura_330d_nylon66 | 0.056 | m | 0.06 yd | 42.9 | 15.3 | 0.85 | same / base | 0.176 | record cordura_330d_nylon66; net 0.059 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags) |
| fabric_alternate | hdpe_sheet: B-07sx2 MS-sx2 | hdpe_sheet | 0.047 | m2 | 0.51 ft2 | 42.9 | 4.58 | 0.21 | same / base,meat_trip | 0.052 | record hdpe_sheet; net 0.033 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags) |
| fabric_alternate | hypalon_coated_nylon: B-17x2 B-18x2 B-22x2 B-11x8 | hypalon_coated_nylon | 0.055 | m2 | 0.59 ft2 | 42.9 | 27.9 | 1.53 | same / base | 0.096 | record hypalon_coated_nylon; net 0.038 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags) |

| Section | Item | Material id | Qty | Unit | Imperial | Waste % | $ / unit | $ | Alternate / kit | Proto qty | Source |
|---|---|---|---|---|---|---|---|---|---|---|---|
| fabric_alternate | nylon_ripstop_70d: B-06x1 B-09cx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1 | nylon_ripstop_70d | 0.892 | m | 0.98 yd | 17.6 | 11.8 | 10.48 | robic_100d_ripstop / addon,as_hunted,base,meat_trip | 1.129 | record nylon_ripstop_70d; net 1.117 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85 |
| fabric_alternate | polyester_ht_630d_printable: B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1 | polyester_ht_630d_printable | 0.158 | m | 0.17 yd | 17.6 | 19.7 | 3.10 | same / addon,as_hunted,base | 0.738 | record polyester_ht_630d_printable; net 0.201 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85 |
| fabric_alternate | power_mesh_nylon_spandex: FP-ax2 | power_mesh_nylon_spandex | 0.129 | m | 0.14 yd | 42.9 | 10.9 | 1.41 | same / base | 0.246 | record power_mesh_nylon_spandex; net 0.136 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags) |
| fabric_alternate | robic_100d_ripstop: IMP4-ax1 IMP4-bx1 | robic_100d_ripstop | 0.115 | m | 0.13 yd | 42.9 | 17.5 | 2.02 | same / base | 0.296 | record robic_100d_ripstop; net 0.123 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags) |
| fabric_alternate | silicone_coated_poly_ripstop_40d: F-01ax1 F-01bx2 MS-ax1 | silicone_coated_poly_ripstop_40d | 1.157 | m | 1.27 yd | 17.6 | 10.9 | 12.65 | same / as_hunted,meat_trip | 1.715 | record silicone_coated_poly_ripstop_40d; net 1.498 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85 |
| fabric_alternate | tpu_coated_nylon_840d: B-12x6 B-09px2 IMP8-bx1 | tpu_coated_nylon_840d | 0.036 | m2 | 0.39 ft2 | 42.9 | 16 | 0.58 | same / base | 0.09 | record tpu_coated_nylon_840d; net 0.025 m2 from the pattern areas; ASSUMPTION UTIL-SMALL 0.70 (small / sheet pieces nested across bags) |
| fabric_alternate | ultragrid_210d_uhmwpe_ripstop: B-03x2 B-02doorx1 B-02hx1 B-07x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1 | ultragrid_210d_uhmwpe_ripstop | 0.75 | m | 0.82 yd | 17.6 | 21.9 | 16.40 | cordura_330d_nylon66 / addon,as_hunted,base | 0.963 | record ultragrid_210d_uhmwpe_ripstop; net 0.956 m2 from the pattern areas; ASSUMPTION UTIL-ALT 0.85 |
| webbing | elastic_25mm: K-28 K-G K-IM K-RT | elastic_25mm | 1.04 | m | 1.14 yd | 5 | 1.1 | 1.14 | same / as_hunted,base | | record elastic_25mm |
| webbing | elastic_6mm: F-EL | elastic_6mm | 2.558 | m | 2.80 yd | 5 | 0.6 | 1.53 | same / as_hunted | | record elastic_6mm |
| webbing | poly_webbing_25mm: B-25 B-28 B-28s B-28t BRL BRS E5 GBL HPL I2K I2T IM1 IM2 IM3 K-W LLT RT TG US W1 W2 W3 WX | poly_webbing_25mm | 16.633 | m | 18.19 yd | 5 | 0.7 | 11.64 | same / as_hunted,base,crown kit M,crown kit M (as hunted),meat_trip | | record poly_webbing_25mm |
| webbing | poly_webbing_38mm: B-24 HE IME T3 | poly_webbing_38mm | 2.825 | m | 3.09 yd | 5 | 1 | 2.82 | same / base,meat_trip | | record poly_webbing_38mm |
| webbing | shock_cord_3mm: B-26 E5-cord FP-cord MK-cord | shock_cord_3mm | 1.847 | m | 2.02 yd | 5 | 0.5 | 0.92 | same / base | | record shock_cord_3mm |
| hardware | #10 coil zip chain, door D2 (2 sliders) | coil_zipper_10 | 1.418 | m | 1.55 yd | 0 | 4 | 5.67 | same / base | | record coil_zipper_10; ledger rows: D2 door zip |
| hardware | #10 coil slider, metal pull (door) | slider_10 | 2 | unit | | 0 | 1.99 | 3.98 | same / base | | record slider_10; ledger rows: D2 door zip |
| hardware | #5 coil zip chain: lid, IMP-4, F-02 (3 zips) | coil_zipper_5 | 0.97 | m | 1.06 yd | 0 | 2 | 1.94 | same / base,as_hunted | | record coil_zipper_5; ledger rows: - |
| hardware | #5 coil slider + pull (3: lid, IMP-4, F-02) | slider_5 | 3 | unit | | 0 | 0.96 | 2.88 | same / base,as_hunted | | record slider_5; ledger rows: BL-034 records: F-02 fly pocket #5 slider |
| hardware | door pull cord 3 mm Dyneema, 2 x 0.15 m (ASSUMPTION length) | dyneema_cord_3mm | 0.3 | m | 0.33 yd | 0 | 0.9 | 0.27 | same / base | | record dyneema_cord_3mm; ledger rows: door: 2 cord pulls |
| hardware | rubber pull sleeve | rubber_pull_sleeve | 2 | unit | | 0 | 0.15 | 0.30 | same / base | | ASSUMPTION; ledger rows: door: 2 cord pulls |
| hardware | rated 25 mm aluminium G-hook: IM1, IM2 (2) + W1-W3 (3) + bow riser (1) | g_hook_25mm_aluminium | 6 | unit | | 0 | 4 | 24.00 | same / base | | record g_hook_25mm_aluminium; ledger rows: BL-010 S9; IM-hooks; W-hw |
| hardware | rated 25 mm aluminium G-hook: W haul extenders | g_hook_25mm_aluminium | 3 | unit | | 0 | 4 | 12.00 | same / meat_trip | | record g_hook_25mm_aluminium; ledger rows: - |
| hardware | wiregate carabiner 22 kN: IM3 bag hooks (2) + IM3 meat free end (1) | carabiner_wiregate_22kn | 3 | unit | | 0 | 10 | 30.00 | same / base | | record carabiner_wiregate_22kn; ledger rows: BL-010 A1; BL-010 S2; attachment system: bag hooks 1500; attachment: bag hooks, 38 mm |

| Section | Item | Material id | Qty | Unit | Imperial | Waste % | $ / unit | $ | Alternate / kit | Proto qty | Source |
|---|---|---|---|---|---|---|---|---|---|---|---|
| hardware | wiregate carabiner 22 kN: B-24 float links | carabiner_wiregate_22kn | 2 | unit | | 0 | 10 | 20.00 | same / meat_trip | | record carabiner_wiregate_22kn; ledger rows: B-24-hooks; BL-010 S12 |
| hardware | heavy 25 mm ladder lock (9): B-28 (4) + guys (2) + W1-W3 (3) | ladder_lock_25mm_heavy | 9 | unit | | 0 | 2.1 | 18.90 | same / base,as_hunted | | record ladder_lock_25mm_heavy; ledger rows: BL-010 A3; BL-010 S9; W-hw |
| hardware | 25 mm aluminium cam buckle: IM1-IM3 (3) + US1/US2 (2) | cam_25mm_al | 5 | unit | | 0 | 25.5 | 127.50 | same / base | | record cam_25mm_al; ledger rows: IM-cams; US-cams |
| hardware | 25 mm STAINLESS tri-glide: US1/US2 cradle backstop behind the cam (2; orchestrator 2026-11-13, 11 g, 3.75 kN) | triglide_25mm | 2 | unit | | 0 | 2.84 | 5.68 | same / base | | record triglide_25mm stainless_alt (EUR 3.10 incl. VAT, converted as the record); ledger rows: BL-034 records: US1/US2 backstop tri-glides; US-tg |
| hardware | 25 mm nylon tri-glide: guy tail keepers (2) | triglide_25mm | 2 | unit | | 0 | 0.37 | 0.74 | same / base | | record triglide_25mm; ledger rows: BL-034 records: guy tri-glides |
| hardware | POM toggle 30 x 8 x 5 (4): B-28 rear (2) + guy loops (2) | pom_toggle | 4 | unit | | 0 | 0.12 | 0.48 | same / base,as_hunted | | record pom_toggle; ledger rows: - |
| hardware | 25 mm side-release buckle (roll top) | buckle_side_release_25mm | 2 | unit | | 0 | 0.8 | 1.60 | same / base | | record buckle_side_release_25mm; ledger rows: - |
| hardware | spring cord lock (3): cord lock (E-5) + rifle muzzle keepers (2, BL-035) | cord_lock_spring | 3 | unit | | 0 | 0.4 | 1.20 | same / base | | record cord_lock_spring; ledger rows: - |
| hardware | TPU sleeve on the IM3 carabiner (2) | tpu_hook_sleeve | 2 | unit | | 0 | 0.2 | 0.40 | same / base | | ASSUMPTION; ledger rows: attachment: TPU hook sleeves |
| hardware_frame_side | printed fairleads (8), 7075 pins (8), UHMWPE sleeves (8), US anchors, lifter keepers | frame kit | 1 | set | | 0 | | | / base | | ledger rows: BL-034 records: lifter tri-glide keepers; US-anchor; attachment: 7075 pins; attachment: UHMWPE sleeves; attachment: lifter tri-glide; attachment: printed fairleads |
| thread | bonded_nylon_tex135 (needle + bobbin) | bonded_nylon_tex135 | 29.1 | m | 31.82 yd | 15 | 0.0081 | 0.24 | same / all | | record bonded_nylon_tex135; sum of op stitch lengths x 2.75 (ASSUMPTION THREAD-FACTOR) + 15% cone waste on the purchase quantity only |
| thread | bonded_nylon_tex70 (needle + bobbin) | bonded_nylon_tex70 | 83.6 | m | 91.43 yd | 15 | 0.0052 | 0.43 | same / all | | record bonded_nylon_tex70; sum of op stitch lengths x 2.75 (ASSUMPTION THREAD-FACTOR) + 15% cone waste on the purchase quantity only |
| thread | bonded_nylon_tex90 (needle + bobbin) | bonded_nylon_tex90 | 124.8 | m | 136.48 yd | 15 | 0.0061 | 0.76 | same / all | | record bonded_nylon_tex90; sum of op stitch lengths x 2.75 (ASSUMPTION THREAD-FACTOR) + 15% cone waste on the purchase quantity only |
| consumable | 6 mm double-sided basting tape (zips) | basting_tape_6mm | 3.66 | m | 4.00 yd | 0 | 0.05 | 0.18 | / | | ASSUMPTION |
| consumable | labels: size / care / ID, hang tag, load-limit warning card | label_set | 1 | set | | 0 | 0.6 | 0.60 | / | | ASSUMPTION: size / care / ID label + hang tag + warning card |
| consumable | silicone seam sealer for the silpoly fly and liner seams, ~10 ml (ASSUMPTION) | seam_sealer | 1 | bag | | 0 | 0.8 | 0.80 | / | | ASSUMPTION: 10 ml of a silicone seam sealer |
| packaging | poly bags + carton share | polybag_carton | 1 | bag | | 0 | 0.9 | 0.90 | / | | ASSUMPTION: 2 recycled poly bags + 1/10 carton |

# 5. Cutting plan (pattern list, markers, cut lengths)

Markers: ../patterns/markers/. Cut windows, slots and slits first; knife clockwise from the grain arrow; notches 6 mm. Ply height: woven nylon 24, laminates / mesh / silpoly 12, camo single ply (one way, print placement), sheet goods laser single sheet.

| Set | Material | Pieces | 1-bag m | eff | 10-bag m / bag | eff | Lay | Prototype | Production |
|---|---|---|---|---|---|---|---|---|---|
| primary | challenge_ultra_200 | B-01x1 B-06fx1 B-13xx6 B-27fx2 | 0.743 | 0.5838 | 0.4726 | 0.9177 | laminate face up, all pieces one way | rotary; windows and B-27 slits with a hot knife / scalpel on a cutting mat | straight knife, 12 plies max (laminate creeps), or laser single ply |
| primary | cordura_330d_nylon66 | B-03x2 B-02doorx1 B-02hx1 B-07x1 IMP8-ax1 F-02x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1 | 1.1348 | 0.6178 | 0.7996 | 0.8767 | face up, one way not needed | rotary + hot-knife the window / slot cut-outs | straight knife, 10-bag marker, 24 plies max; or laser single ply (sealed edges) |
| primary | cordura_500d_nylon66 | B-05x1 B-16x2 | 0.22 | 0.3255 | 0.0889 | 0.8055 | face up | rotary | straight knife, 24 plies max |
| primary | hdpe_sheet | B-07sx2 MS-sx2 | 0.043 | 0.6213 | | | free rotation | shear / router, round every corner R >= 3 | CNC router from the DXF |
| primary | hypalon_coated_nylon | B-17x2 B-18x2 B-22x2 B-11x8 | 0.07 | 0.4063 | | | coated face up, free rotation | laser from the DXF (makerspace) or a knife + template | laser single sheet; dies above 500 bags |
| primary | polyester_ht_630d_printable | B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1 | 0.738 | 0.1878 | 0.2961 | 0.468 | print face up, ONE WAY | rotary | single ply (printed camo, one way, pattern placement) |
| primary | power_mesh_nylon_spandex | FP-ax2 | 0.246 | 0.3807 | | | stretch ACROSS, one way | rotary, no stretch while cutting | straight knife 12 plies, relaxed 24 h before cutting |
| primary | robic_100d_ripstop | B-06x1 B-09cx1 IMP4-ax1 IMP4-bx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1 | 1.208 | 0.6966 | 0.9305 | 0.9043 | face up | rotary | straight knife, 24 plies max (paper between every 6 plies: slippery) |
| primary | silicone_coated_poly_ripstop_40d | F-01ax1 F-01bx2 MS-ax1 | 1.7146 | 0.5932 | | | face up | rotary, weights not pins | straight knife 12 plies with paper underlay |
| primary | tpu_coated_nylon_840d | B-12x6 B-09px2 IMP8-bx1 | 0.06 | 0.2887 | | | coated face up, free rotation | laser from the DXF | laser single sheet |
| alternate | challenge_ultra_200 | B-01x1 B-05x1 B-06fx1 B-13xx6 B-27fx2 B-16x2 | 0.743 | 0.6837 | | | laminate face up, all pieces one way | rotary; windows and B-27 slits with a hot knife / scalpel on a cutting mat | straight knife, 12 plies max (laminate creeps), or laser single ply |
| alternate | cordura_330d_nylon66 | IMP8-ax1 F-02x1 | 0.176 | 0.2327 | | | face up, one way not needed | rotary + hot-knife the window / slot cut-outs | straight knife, 10-bag marker, 24 plies max; or laser single ply (sealed edges) |
| alternate | hdpe_sheet | B-07sx2 MS-sx2 | 0.043 | 0.6213 | | | free rotation | shear / router, round every corner R >= 3 | CNC router from the DXF |
| alternate | hypalon_coated_nylon | B-17x2 B-18x2 B-22x2 B-11x8 | 0.07 | 0.4063 | | | coated face up, free rotation | laser from the DXF (makerspace) or a knife + template | laser single sheet; dies above 500 bags |
| alternate | nylon_ripstop_70d | B-06x1 B-09cx1 B-29bx1 A1-bx1 MS-bx1 MS-cx1 | 1.129 | 0.6714 | | | face up | rotary | straight knife, 12 plies max (slippery) |
| alternate | polyester_ht_630d_printable | B-02ux1 SH-07x1 SH-01bx2 FP-bx2 B-29fx2 A1-dx1 | 0.738 | 0.1878 | | | print face up, ONE WAY | rotary | single ply (printed camo, one way, pattern placement) |
| alternate | power_mesh_nylon_spandex | FP-ax2 | 0.246 | 0.3807 | | | stretch ACROSS, one way | rotary, no stretch while cutting | straight knife 12 plies, relaxed 24 h before cutting |
| alternate | robic_100d_ripstop | IMP4-ax1 IMP4-bx1 | 0.296 | 0.2819 | | | face up | rotary | straight knife, 24 plies max (paper between every 6 plies: slippery) |
| alternate | silicone_coated_poly_ripstop_40d | F-01ax1 F-01bx2 MS-ax1 | 1.7146 | 0.5932 | | | face up | rotary, weights not pins | straight knife 12 plies with paper underlay |
| alternate | tpu_coated_nylon_840d | B-12x6 B-09px2 IMP8-bx1 | 0.06 | 0.2887 | | | coated face up, free rotation | laser from the DXF | laser single sheet |
| alternate | ultragrid_210d_uhmwpe_ripstop | B-03x2 B-02doorx1 B-02hx1 B-07x1 B-29ax1 B-29cx1 B-29dx1 B-29ex1 B-29gx2 A1-ax1 | 0.963 | 0.6729 | | | face up | rotary (UHMWPE grid: new blade) | straight knife 12 plies; UHMWPE dulls blades: sharpen every lay |

# Webbing, cord and elastic cut lengths

| ID | Use | Material | Length mm (in) | Qty | Kit | Source |
|---|---|---|---|---|---|---|
| B-28 | B-28 crown strap, kit S | poly_webbing_25mm | 1068 (42.1) | 2 | crown kit S (as hunted) | corrie_f.crown_kit_straps |
| B-28 | B-28 crown strap, kit M | poly_webbing_25mm | 1007 (39.6) | 2 | crown kit M (as hunted) | corrie_f.crown_kit_straps |
| B-28 | B-28 crown strap, kit L | poly_webbing_25mm | 948 (37.3) | 2 | crown kit L (as hunted) | corrie_f.crown_kit_straps |
| B-28t | lock tab | poly_webbing_25mm | 60 (2.4) | 4 | as_hunted | corrie_f.CROWN_TAB, 2 per strap (front + rear lock) |
| B-28s | toggle loop | poly_webbing_25mm | 60 (2.4) | 2 | as_hunted | corrie_f.CROWN_TAB, 1 per strap (the B-25 second fold) |
| K-28 | B-28 tail keeper (25 mm elastic loop) | elastic_25mm | 90 (3.5) | 4 | as_hunted | corrie_f.crown_kit_straps keepers_per_strap x 2 |
| W1 | haul wrap W1 (cut for carry) | poly_webbing_25mm | 750 (29.5) | 1 | base | strap_tails.wrap_rows |
| W2 | haul wrap W2 (cut for carry) | poly_webbing_25mm | 822 (32.4) | 1 | base | strap_tails.wrap_rows |
| W3 | haul wrap W3, kit S | poly_webbing_25mm | 1034 (40.7) | 1 | crown kit S | strap_tails.wrap_rows |
| W3 | haul wrap W3, kit M | poly_webbing_25mm | 987 (38.8) | 1 | crown kit M | strap_tails.wrap_rows |
| W3 | haul wrap W3, kit L | poly_webbing_25mm | 935 (36.8) | 1 | crown kit L | strap_tails.wrap_rows |
| WX | W haul extender (meat-trip kit) | poly_webbing_25mm | 740 (29.1) | 3 | meat_trip | strap_tails.wrap_rows haul extender_mm |
| K-W | W gore keeper loop (6, in the gore seams) | poly_webbing_25mm | 80 (3.1) | 6 | base | ledger 'keepers: 6 haul-wrap keeper loops' (ASSUMPTION 80 mm cut) |
| RT | roll-top side clip strap | poly_webbing_25mm | 510 (20.1) | 2 | base | strap_tails.roll_row |
| K-RT | roll clip tail keeper (25 mm elastic) | elastic_25mm | 90 (3.5) | 2 | base | BL-031 |
| IM1 | internal meat strap IM1 | poly_webbing_25mm | 990 (39.0) | 1 | base | strap_tails.im_rows (unchanged) |
| IM2 | internal meat strap IM2 | poly_webbing_25mm | 1023 (40.3) | 1 | base | strap_tails.im_rows (unchanged) |
| IM3 | internal meat strap IM3 | poly_webbing_25mm | 1083 (42.6) | 1 | base | strap_tails.im_rows (unchanged) |
| K-IM | IM tail keeper (25 mm elastic) | elastic_25mm | 90 (3.5) | 3 | base | BL-031 |
| IME | IM fairlead eye (38 mm) | poly_webbing_38mm | 200 (7.9) | 6 | base | ledger IM-eyes 6 x 0.20 m |
| US | under-strap US1 / US2 | poly_webbing_25mm | 550 (21.7) | 2 | base | ledger US-web 1.10 m |
| B-24 | float stand-off link (38 mm, meat-trip) | poly_webbing_38mm | 435 (17.1) | 2 | meat_trip | attach_system.link_piece; ledger B-24-web 2 x 435 mm |
| HE | hinge eye (38 mm) | poly_webbing_38mm | 150 (5.9) | 2 | base | attach_system.drawing_pieces HE |
| T3 | IM3 bag-hook tongue (38 mm) | poly_webbing_38mm | 160 (6.3) | 2 | base | attach_system.drawing_pieces T2/T3 (IM2 hooks removed by BL-004: 2 left) |
| B-25 | guy bridle on the crown top bar | poly_webbing_25mm | 412 (16.2) | 1 | base | corrie_f.guy_bridle |
| TG | port guy (B-25 toggle -> B-27 port -> B-22 inside) | poly_webbing_25mm | 500 (19.7) | 2 | base | attach_system.GUY cut_mm |
| LLT | guy ladder-lock tab on B-22 | poly_webbing_25mm | 70 (2.8) | 2 | base | attach_system.drawing_pieces LLT |
| K-G | guy tail keeper (25 mm elastic) | elastic_25mm | 90 (3.5) | 2 | base | ledger 'guy tail elastic keepers x2' |
| GBL | guy bar loop (POM toggle) | poly_webbing_25mm | 170 (6.7) | 2 | base | ledger BL-004 REC 'guy bar loops x2: 25 mm x 170 mm' |
| HPL | scope-pod hard-point loop | poly_webbing_25mm | 80 (3.1) | 4 | base | corrie_f.LOOP_M |
| E5 | pole grip tab (cord lock) | poly_webbing_25mm | 60 (2.4) | 1 | base | evolve.POLE tab_m |
| I2T | IMP-2 hang tab (B-06f) | poly_webbing_25mm | 100 (3.9) | 1 | base | lean.fell IMP-2L 0.10 m |
| I2K | IMP-2 bottom keeper tab (B-01) | poly_webbing_25mm | 60 (2.4) | 2 | base | lean.fell IMP-2L 0.06 m |
| BRL | bow riser loop (through B-11) | poly_webbing_25mm | 100 (3.9) | 2 | base | ledger bow riser carry (ASSUMPTION 100 mm loops) |

| ID | Use | Material | Length mm (in) | Qty | Kit | Source |
|---|---|---|---|---|---|---|
| BRS | bow riser strap + G-hook | poly_webbing_25mm | 300 (11.8) | 1 | base | ledger bow riser carry (ASSUMPTION 300 mm strap) |
| B-26 | B-26 tripod keeper, 3 mm shock cord (ends 15 mm in the R corner seams) | shock_cord_3mm | 201 (7.9) | 1 | base | corrie_f.tripod_keeper |
| FP-cord | FP mouth shock cord | shock_cord_3mm | 320 (12.6) | 2 | base | corrie_f.hip_reach_pockets (w + pleat + 60) |
| E5-cord | pole quiver grip cord | shock_cord_3mm | 300 (11.8) | 1 | base | evolve.POLE |
| MK-cord | rifle muzzle keeper, 3 mm shock cord loop through the 2 B-11 muzzle tabs + spring cord lock (one per gusset) | shock_cord_3mm | 309 (12.2) | 2 | base | corrie_f.muzzle_keeper (ASSUMPTION MUZZLE-CORD) |
| F-EL | fly hem elastic, 6 mm (spine top + bottom, sides bottom + frame) | elastic_6mm | 2436 (95.9) | 1 | as_hunted | corrie_f.fly_pieces (FLY_ELASTIC_RATIO x hem channels) |

# 6. Sewing operation sequence

All flat work before the bag becomes 3D. Clips, never pins, in coated fabric. Swatch every new stack. Load points only after their proof test passes (section 8). V = volatile (re-run on a pattern change).

| Seq | Operation | How (beginner) | Pieces | Machine | Needle | Thread | SPI | Stitch / seam | mm | Ref | h / SAM min |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.01 | Print and check the test square | Print patterns/print_letter/00_PRINT_FIRST_test_square.pdf at 100 %. The solid square must read 100 mm (+/- 1 mm) both ways, the dashed one 4 in. Do not cut from a wrong-scale print. | | bench | - | - | - | - / - | | QC-01 | 0.10 / 0.0 1x |
| 0.02 | Machine set-up and stack swatches | Make a 150 x 150 mm swatch of each stack and set tension until the lock sits mid-stack on both faces: (a) ULTRA + 330D body seam, tex 90 / Nm 110 R, 8 SPI; (b) Hypalon on ULTRA, double row; (c) 25 mm box-X tex 135 / Nm 140 R, 7 SPI; (d) 70D / mesh at 10 SPI. Keep them in the QC folder. | | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa2, LSd, BOX, SSa (swatches) | | SK-1, SK-2; QC-05, QC-06 | 1.00 / 0.0 1x |
| 0.03 | Sew the first-article load samples | Before cutting the bag, sew the proof specimens exactly as the bag: 10 x 25 mm box-X loops (MT-1), 5 T3 tongues on 70 mm B-17 tabs on B-01 swatches (PT-2), 5 HE eyes on B-18 + B-01 + B-05 swatches without binding (PT-3), 5 B-24 links (PT-6). The bag's load points are sewn only after these pass. | | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX + LSd (as the bag) | | SK-1; TEST gates; all load points | 3.00 / 0.0 1x |
| 0.04 | Cut 77 fabric / sheet pieces | Per cutting_plan.csv: lay face up (one way for camo and mesh), L+R pieces face to face, cut windows, slots and slits FIRST, knife clockwise from the grain arrow. Notches 6 mm, never past the seam line. Hypalon, TPU-840D and HDPE from the DXF on the laser / router. | all | SK / LASER / HK | - | - | - | - / - | | QC-02, QC-03, QC-04 | 9.24 / 66.5 |
| 0.05 | Hot-knife 75 webbing / elastic / cord pieces | Cut lengths in cutting_plan.csv (crown kit M shown; B-28 and W3 per crown kit). Use the J8 stop; seal both ends; no hard bead wider than the webbing. | webbing | SK / LASER / HK | - | - | - | - / - | | QC-04 | 2.25 / 16.2 |
| 0.06 | Mark every placement | Chalk / silver pen on the INSIDE unless the pattern says OUTSIDE; awl the drill marks (box corners, slit ends, window corners, zip stops). Pieces: B-01, B-03 (L only / R only marks), B-02h, B-02door, B-02u, B-06f, B-29a. | | bench | - | - | - | - / - | | | 0.60 / 4.3 |
| 1.01 V | FP hip-reach pocket: pleats | Fold the 2 box pleats on each FP-a (flat 260 mm wide, finished 165 mm), baste across top and bottom at 6 mm. Do not stretch the mesh. | FP-a x2 | WF1 | Nm 90/14 DPx17, medium ball SUK (knit mesh) | BN tex70 | 10 | 301 / SSa | 1040 | SK-5 | 1.02 / 7.4 |
| 1.02 V | FP mouth band and shock-cord channel | Sew FP-b to the FP-a top (260 mm), fold the band on its fold line over the seam, edge-stitch 2 mm to form the channel; thread the shock cord (FP-cord) and bar it at both ends inside the side allowance. | FP-a, FP-b x2, FP-cord | WF1 | Nm 90/14 DPx17, medium ball SUK (knit mesh) | BN tex70 | 10 | 301 / SSa + EFa channel | 1040 | SK-2 | 0.62 / 4.5 |
| 1.03 | IMP-4 zip pocket | Set the #5 zip (210 mm) in the top of IMP4-a (basting tape, ZIP seam), stops at the drill marks; sew the bellows strip IMP4-b round bottom + sides of IMP4-a. | IMP4-a, IMP4-b, #5 zip | WF1 | Nm 90/14 DPx17, round point R | BN tex70 | 10 | 301 / ZIP + SSa | 1240 | | 2.29 / 16.5 |
| 1.04 | IMP-8 cam cup | Fold IMP8-a, sew the 2 side seams; sew the laser-cut TPU base IMP8-b in (TPU face out). | IMP8-a, IMP8-b | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa | 610 | | 0.75 / 5.4 |
| 1.05 | IMP-2 hydration sleeve | Fold B-09c into a tube, one side seam; turn the top edge 10 mm and edge-stitch; I2T hang tab goes in later (op 5.01). | B-09c | WF1 | Nm 90/14 DPx17, round point R | BN tex70 | 10 | 301 / SSa + EFa | 930 | | 0.56 / 4.0 |
| 1.06 V | F-02 fly base pocket | Set the #5 zip (230 mm), turn the edges 10 mm. It is sewn to the floor B-05 in step 4 on the drawn 'F-02 fly pocket' placement (190 x 140 mm, OUTSIDE, zip on the aft edge, corners drilled; pattern ASSUMPTION F02-CENTRE fore-aft). | F-02, #5 zip | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / ZIP + EFa | 1120 | | 1.13 / 8.1 |
| 2.01 | B-12 strap-port patches on the swell crests | Lay a B-12 (TPU face out) on the crest of each B-13x (6, not the hinge swells), sew the 3 mm perimeter, then cut the 45 x 12 slot through both and edge-stitch the slot. | B-12 x6, B-13x x6 | WF2 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / LSd | 1843 | SK-2 | 1.47 / 10.6 |

| Seq | Operation | How (beginner) | Pieces | Machine | Needle | Thread | SPI | Stitch / seam | mm | Ref | h / SAM min |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2.02 V | Box the swells | Sew the 4 corner folds of every swell (B-13x x6): fold on 'box the corner', sew across. Practice: set one swell into a scrap window before touching B-01. | B-13x x6 | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa | | | 1.50 / 10.8 |
| 2.03 | T3 IM3 tongues onto B-17 | Fold each T3 45 mm round the bed (empty loop: the carabiner clips in later), box-X the loop; box-X the other end to the centre of a B-17 (Hypalon, coated face out). | T3 x2, B-17 x2 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 902 | SK-1; QC-07; T3 (load table) | 0.40 / 2.9 |
| 2.04 | IME fairlead eyes (frame kit) | Fold each 38 mm IME at 100 mm into an eye, box-X 34 x 30. These go on the frame fairleads, not the bag. | IME x6 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 1312 | QC-07; IME | 0.60 / 4.3 |
| 3.01 | SH-07 slider garage on the right blade | Turn the edges 8 mm and topstitch SH-07 to B-02u at its placement. | SH-07, B-02u | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / LSd | 200 | | 0.16 / 1.1 |
| 3.02 | IMP-4 onto the door, INSIDE face | Sew the IMP-4 bellows edge to B-02door at the IMP-4 placement (inside), pivot at the drill marks. | IMP-4, B-02door | WF1 | Nm 90/14 DPx17, round point R | BN tex70 | 10 | 301 / SSa | 820 | | 0.69 / 5.0 |
| 3.03 | Door #10 zip | Basting-tape the #10 zip to the door edge and to the blade U inner edge (1358 mm, 53.5 in), notch to notch, corner tangent notches at each R 55 corner; 2 sliders head to head; sew each tape at 5 mm, turn, edge-stitch 3 mm; zip stops at the hinge band. | B-02door, B-02u, #10 zip | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / ZIP | 5432 | SK-2; QC-09 | 2.66 / 19.1 |
| 3.04 V | B-22 guy anchors INSIDE the hinge band | Place each B-22 (coated face down on the band's inside) at x +/-130.3 mm (5.13 in) on the drawn placement (drawn x +/-149, moved 18.7 mm inboard, ASSUMPTION B22-IN). Double-row perimeter 3 / 9 mm. | B-22 x2, B-02h | WF2 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / LSd (2 rows) | 839 | SK-4; QC-07 | 0.67 / 4.8 |
| 3.05 V | Guy ladder-lock tabs LLT through B-22 | Fold each LLT round the bar of a heavy 25 mm ladder lock, box-X through LLT + B-22 + B-02h at the drawn 'B-22 / guy LLT box' (drill corners). Box 21 x 25 mm (LLT piece). | LLT x2, B-22, B-02h, 2 heavy ladder locks | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 315 | SK-1, SK-4; QC-07; LLT + B-22 + B-02h | 0.20 / 1.4 |
| 3.06 | B-28 front tabs on the hinge band (OUTSIDE) | Fold a B-28t round the bar of a heavy ladder lock (lock bar on the top line of the drawn 'B-28 front lock tab' placement), box-X it through tab + band on the OUTSIDE of B-02h at the drawn 'B-28 front tab (OUTSIDE)' box, 21 x 20 mm. DECISION B28-FACE (BL-034): the crown strap runs down the band's aft face to the lock, so the tab is outside. | B-28t x2, B-02h, 2 heavy ladder locks | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 280 | SK-4; QC-07; B-28t | 0.20 / 1.4 |
| 3.07 | Hinge band to the door top | Join B-02h to the door top (334 mm), SSa2. | B-02h, B-02door | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa2 | 668 | | 0.20 / 1.4 |
| 3.08 | Hinge band ends to the blade U | Join both short band ends to the blade tops (193 mm total). | B-02h, B-02u | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa2 | 385 | | 0.22 / 1.6 |
| 4.01 V | Check the B-01 cut-outs | 6 swell windows, 2 B-27 slits (156 mm at x +/-52), IMP-2 keeper drill marks. All cut, corners awled. | B-01 | bench | - | - | - | - / - | | QC-03 | 0.10 / 0.7 |
| 4.02 V | Set the 6 swells into the B-01 windows | Swell base to window edge, INSIDE B-01, notch to notch (6 windows, 1680 mm of window edge in all; hinge windows: 0). Sew from the swell side, needle down at every corner. | B-13x x6, B-01 | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa | 1680 | SK-3; QC-08 | 1.50 / 10.8 |
| 4.03 | B-27 guy ports (2) | Each slit (156 mm, x +/-52, z 595-745): turn both edges 6 mm and edge-stitch; sew the B-27f facing INSIDE along the inboard stitch line so the outboard edge laps 20 mm over it; bartack both slit ends. It crosses the setback crease at z 631: the facing follows the crease. | B-01, B-27f x2 | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / EFa + LSd (lapped port) | 965 | SK-2, SK-3; QC-10; B-27 slit ends | 0.98 / 7.0 |
| 4.04 | B-17 IM3 tabs OUTSIDE B-01 | Place each B-17 (with its T3) OUTSIDE B-01 at x +/-115, z 385-455 (placement). Double-row perimeter 3 / 9 mm, 8 SPI, tex 90; the 90 mm top edge carries the load: no gaps, backtack 10 mm. | B-17 x2, B-01 | WF2 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / LSd (2 rows) | 1239 | SK-3; QC-07; B-17 perimeter | 0.99 / 7.1 |

| Seq | Operation | How (beginner) | Pieces | Machine | Needle | Thread | SPI | Stitch / seam | mm | Ref | h / SAM min |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 4.05 V | B-18 doublers + HE hinge eyes | Lay each B-18 INSIDE B-01 at the placement (lower edge to the floor seam line). Fold the HE eye (50 mm back, 25 mm eye; first 30 mm tri-folded to an 18 mm neck, 304 bartack across the neck), box-X 34 x 34 through HE + B-18 + B-01, then the B-18 perimeter rows on 3 sides (the 4th goes in the floor seam). | HE x2, B-18 x2, B-01 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX + LSd | 1243 | SK-1, SK-3; QC-07; HE + B-18 + B-01 | 1.02 / 7.4 |
| 4.06 | IMP-2 keeper tabs I2K on B-01 | Fold each I2K, bartack at the 2 'IMP-2 keeper tab' drill marks (inside). | I2K x2, B-01 | BT | Nm 110/18 DPx17, round point R | BN tex90 | - | 304 / BT | | | 0.20 / 1.4 |
| 4.07 | B-11 lash tabs (8) on the gussets | Gusset flat work, before the gusset seams: 4 tabs on each B-03 at the drawn 'B-11 lash tab' placements (bow riser / lash z 470 (175 mm aft of the frame seam, 40 mm forward of the aft corner); bow riser / lash z 560 (186 mm aft of the frame seam, 40 mm forward of the aft corner); muzzle keeper z 650 (60 mm aft of the frame seam); muzzle keeper z 650 (130 mm aft of the frame seam)). Tab coated face out, slot end UP; awl the 4 'B-11 box corner' drill marks; sew the 24 x 19 mm 'B-11 sew-down box' (rectangle + both diagonals, 301) through tab + gusset. Right gusset: bow riser; left: antler / cape lash; muzzle keepers both sides (pattern ASSUMPTION B11-PLACE, settled by the archery and rifle loadout fits). | B-11 x8, B-03 x2 | WF2 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / LSd (box-X sew-down) | 1178 | SK-5; QC-07 | 0.80 / 5.8 |
| 4.08 | Gusset flat work: kick bands | SH-01b on each gusset foot at the 'kick band top' line, top edge turned 8 mm, topstitched; bottom and sides go in the seams. | SH-01b x2, B-03 x2 | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / LSd | 320 | | 0.26 / 1.9 |
| 4.09 V | Gusset flat work: FP pockets (both) | Sew each FP to its gusset at the placement, z 150-380, 165 mm face; inner edge >= 9 mm off the frame seam (9.5 mm at the tightest point, pattern fp_min_margin_mm). Bottom and sides 301, box pleats laid flat, bartack the 2 mouth corners. | FP x2, B-03 x2 | WF1 | Nm 90/14 DPx17, medium ball SUK (knit mesh) | BN tex70 | 10 | 301 / LSd | 1250 | SK-5; FP corners (BT-S) | 1.15 / 8.3 |
| 4.10 | Gusset flat work: IMP-8 (R only) and hose ports | IMP-8 cup at 'IMP-8 cup top (R only)'. Hose slots (both gussets, z 700): B-09p binding (TPU, laser-cut) laid over the slot, perimeter + slot edge stitched. | IMP-8, B-09p x2, B-03 x2 | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / LSd | 1090 | | 0.81 / 5.9 |
| 4.11 | Frame-side box seams B-01 \| gussets | R (720 mm): E5 pole-grip tab at z 640 and the B-26 shock-cord ends (15 mm in) inserted. L (720 mm): 2 HPL hard-point loops at z 420 / 640 inserted. Notch to notch every 50 mm; SSa2. | B-01, B-03 x2, E5, E5-cord, B-26, HPL x2 | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa-ins + SSa2 | 2881 | SK-5; QC-08 | 1.16 / 8.4 |
| 4.12 | Aft corner seams blade U \| gussets (closes the tube) | R and L (715 / 715 mm): 2 HPL loops in the L seam at the hard-point notches; 3 K-W keeper loops in each seam, raw ends in the seam, at the W1-W3 crossings: R seam z 128 / 300 / 703; L seam z 202 / 370 / 703 (W1, W2, W3; the drawn 'K-W keeper' notches on B-03 and B-02u, pattern ASSUMPTION KW-PATH). SSa2. | B-02u, B-03 x2, HPL x2, K-W x6 | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa-ins + SSa2 | 2861 | SK-5; QC-08 | 1.66 / 11.9 |
| 4.13 V | US floor channels B-16 on F-02 | Turn both ends of each B-16 15 mm (left OPEN: the 25 mm under-strap passes through), lay it across F-02's outer face on the drawn 'B-16 channel' lines (stations [48.0, 104.0] mm), topstitch both long edges, bartack the 4 corners. | B-16 x2, F-02 | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / LSd | 760 | | 1.01 / 7.2 |
| 4.14 | Floor B-05 (F-02 first) | Sew F-02 to the OUTSIDE of B-05 on the drawn 'F-02 fly pocket' placement (zip edge aft, corner drill marks), perimeter row. Then the floor ring (1107 mm): B-01 edge first with the HE tails and the B-18 lower edges caught in it, then gussets, then the blade kick strip; quarter notches matched, needle down at the 4 corners. SSa2. US channels: on F-02 (op above). | B-05, F-02, B-01, B-03 x2, B-02u | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa2 | 2874 | QC-08 | 1.89 / 13.6 |
| 5.01 | Collar tube | B-06f (ULTRA) to B-06 both sides (990 mm); I2T hang tab at the B-06f drill mark. | B-06f, B-06, I2T | WF1 | Nm 90/14 DPx17, round point R | BN tex70 | 10 | 301 / SSa2 | 1979 | | 0.69 / 5.0 |
| 5.02 | Roll-top hem with HDPE strips and clip straps | Close the hem back seam (70 mm). Sew the hem to the collar top ring (1050 mm), fold over the two B-07s HDPE strips and stitch closed (EFb). RT clip straps (510 mm) at each roll end: 60 mm sewn to the roll end, 30 mm round the side-release buckle bar, box-X. | B-07, B-07s x2, B-06, RT x2, 2 SR buckles | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa + EFb + BOX | 2484 | SK-2 | 1.37 / 9.9 |

| Seq | Operation | How (beginner) | Pieces | Machine | Needle | Thread | SPI | Stitch / seam | mm | Ref | h / SAM min |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 5.03 | Collar ring to the body | Collar to the body tops (1050 mm), matched at every panel-corner V-notch. The seam closes the top end of each B-27 port; bartack the port top over it. | collar, body | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa2 | 2100 | QC-08, QC-10 | 1.33 / 9.6 |
| 6.01 | Lid: 4 crown darts sewn + pressed | Fold each B-29a corner dart on its centre, sew leg to leg from the edge to the apex drill mark, tie off (no backtack at the apex), press the dart toward the aft edge. (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / DART | | SK-6 | 1.00 / 7.2 |
| 6.02 | Lid: strap channels topstitched to the crown, both edges (2 x 2 runs) | Lay each B-29g on its 'strap channel CL' line (x +/-52), edges turned 8 mm, topstitch both long edges; channels are 30 mm wide for the 25 mm B-28 strap (drawn so). (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / LSd | 1062 | SK-6 | 0.64 / 4.6 |
| 6.03 | Lid: bartacks at the 4 channel ends | BT-M bartack across each channel end (tex 90, 304), 3 mm in from the turned end. (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BT | | SK-6 | 0.40 / 2.9 |
| 6.04 | Lid: camo caps: short ends turned and topstitched (2 x 2) | Turn the short ends of each B-29f 10 mm and topstitch the cap on B-29e at its placement; long edges go into the ring seams. (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / LSd | 60 | SK-6 | 0.05 / 0.4 |
| 6.05 | Lid: #5 coil zip: tape baste, stops, slider | Baste the #5 zip between B-29d (upper) and B-29e (lower) with 6 mm tape; zip stops at the drill marks; slider on before closing. (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / ZIP | | SK-6 | 0.50 / 3.6 |
| 6.06 | Lid: #5 coil zip sewn, both tapes | ZIP seam: each tape at 5 mm, turn, edge-stitch 3 mm. (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / ZIP | 940 | SK-6 | 0.47 / 3.4 |
| 6.07 | Lid: zip round the two aft corners | Ease the zip round each aft corner; clip the tape 3 mm at the corner tangent notches. (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / ZIP | | SK-6 | 0.80 / 5.8 |
| 6.08 | Lid: wall ring closed: two short seams | Join the front wall run B-29c to the aft strips at both ends (30 mm seams), SSa2. (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa2 | 120 | SK-6 | 0.04 / 0.3 |
| 6.09 | Lid: crown \| wall perimeter (curved, notch match, bind) | Crown to wall ring notch to notch, darts matched to the corner notches; clip the crown allowance every 15 mm on the curves; crown on top. (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa2 | 1978 | SK-6 | 1.78 / 12.8 |
| 6.10 | Lid: crown \| wall: closed ring handling | Sew the ring in one pass, needle down at each notch; open the zip first. (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa2 | | SK-6 | 0.50 / 3.6 |
| 6.11 | Lid: wall \| base perimeter (curved, notch match, bind), turned through the zip | Base B-29b to the wall ring, notch to notch, inside out; then turn the lid right side out through the open zip. (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | WF1 | Nm 90/14 DPx17, round point R | BN tex70 | 10 | 301 / ZIP | 1978 | SK-6 | 1.78 / 12.8 |
| 6.12 | Lid: wall \| base: closed ring handling | As the crown ring: one pass, needle down at each notch. (rows from patterns_corrie_f_rev3.lid_hours) | B-29a, B-29b, B-29c, B-29d, B-29e, B-29f, B-29g | WF1 | Nm 90/14 DPx17, round point R | BN tex70 | 10 | 301 / SSa2 | | SK-6 | 0.50 / 3.6 |
| 7.01 | B-28: B-28 crown strap, kit M x2 | straight strap, sealed ends; threaded at assembly (op 7.end). | B-28 x2 | bench | - | - | - | - / - | | SK-1, SK-7 | 0.00 / 0.0 |
| 7.02 | B-28t: lock tab (the other 2 are boxed in an earlier op) x2 | 60 mm (2.4 in). Folds: fold round the lock bar. Box-X: 21 x 20 at the drawn marks. | B-28t x2 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 280 | SK-1, SK-7; QC-07; B-28t | 0.20 / 1.4 |
| 7.03 | B-28s: toggle loop x2 | 60 mm (2.4 in). Folds: fold: toggle loop. Box-X: 21 x 15 at the drawn marks. | B-28s x2 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 247 | SK-1, SK-7; QC-07; B-28s | 0.20 / 1.4 |
| 7.04 | K-28: B-28 tail keeper (25 mm elastic loop) x4 | Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes). | K-28 x4 | ZZ | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 304 / BT (zigzag lap) | | | 0.40 / 2.9 |

| Seq | Operation | How (beginner) | Pieces | Machine | Needle | Thread | SPI | Stitch / seam | mm | Ref | h / SAM min |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 7.05 | W1: haul wrap W1 (cut for carry) x1 | 750 mm (29.5 in). Folds: G-hook fold. Box-X: 21 x 25 at the drawn marks. | W1 x1 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 157 | SK-1, SK-7; QC-07; W1 | 0.10 / 0.7 |
| 7.06 | W2: haul wrap W2 (cut for carry) x1 | 822 mm (32.4 in). Folds: G-hook fold. Box-X: 21 x 25 at the drawn marks. | W2 x1 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 157 | SK-1, SK-7; QC-07; W2 | 0.10 / 0.7 |
| 7.07 | W3: haul wrap W3, kit M x1 | 987 mm (38.8 in). Folds: G-hook fold. Box-X: 21 x 25 at the drawn marks. | W3 x1 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 157 | SK-1, SK-7; QC-07; W3 | 0.10 / 0.7 |
| 7.08 | WX: W haul extender (meat-trip kit) x3 | 740 mm (29.1 in). Folds: G-hook fold; sewn loop fold (takes the W's own G-hook). Box-X: 21 x 25, 21 x 25 at the drawn marks. | WX x3 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 944 | SK-1, SK-7; QC-07; WX | 0.60 / 4.3 |
| 7.09 | K-W: W gore keeper loop (6, in the gore seams) x6 | 80 mm (3.1 in). Folds: fold, raw ends in the seam. Box-X / bartack per the load table where it is caught in a seam. | K-W x6 | BT | Nm 110/18 DPx17, round point R | BN tex90 | - | 304 / BT | | SK-1, SK-7 | 0.60 / 4.3 |
| 7.10 | RT: roll-top side clip strap x2 | 510 mm (20.1 in). Folds: sewn to the roll end (60); round the buckle bar (30). Box-X / bartack per the load table where it is caught in a seam. | RT x2 | BT | Nm 110/18 DPx17, round point R | BN tex90 | - | 304 / BT | | SK-1, SK-7 | 0.20 / 1.4 |
| 7.11 | K-RT: roll clip tail keeper (25 mm elastic) x2 | Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes). | K-RT x2 | ZZ | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 304 / BT (zigzag lap) | | | 0.20 / 1.4 |
| 7.12 | IM1: internal meat strap IM1 x1 | 990 mm (39.0 in). Folds: left end loop (girth-hitched on the eye). Box-X: 21 x 25 at the drawn marks. | IM1 x1 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 157 | SK-1, SK-7; QC-07; IM1 | 0.10 / 0.7 |
| 7.13 | IM2: internal meat strap IM2 x1 | 1023 mm (40.3 in). Folds: left end loop (girth-hitched on the eye). Box-X: 21 x 25 at the drawn marks. | IM2 x1 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 157 | SK-1, SK-7; QC-07; IM2 | 0.10 / 0.7 |
| 7.14 | IM3: internal meat strap IM3 x1 | 1083 mm (42.6 in). Folds: left end loop (girth-hitched on the eye). Box-X: 21 x 25 at the drawn marks. | IM3 x1 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 157 | SK-1, SK-7; QC-07; IM3 | 0.10 / 0.7 |
| 7.15 | K-IM: IM tail keeper (25 mm elastic) x3 | Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes). | K-IM x3 | ZZ | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 304 / BT (zigzag lap) | | | 0.30 / 2.2 |
| 7.16 V | US: under-strap US1 / US2 x2 | 550 mm (21.7 in). Folds: eye fold (round the left pin sleeve). Box-X: 21 x 25 at the drawn marks. | US x2 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 315 | SK-1, SK-7; QC-07; US | 0.20 / 1.4 |
| 7.17 | B-24: float stand-off link (38 mm, meat-trip) x2 | 435 mm (17.1 in). Folds: fold round the hook bed (finished 0); loop 1 start; loop 1 apex: gap 120 mm (4.7 in) - T3 hook sits here; loop 1 end (meets start); loop 2 start; loop 2 apex: gap 170 mm (6.7 in) - T3 hook sits here; loop 2 end (meets start); loop 3 start; loop 3 apex: gap 220 mm (8.7 in) - T3 hook sits here; loop 3 end (meets start). Box-X: 34 x 30, 34 x 20, 34 x 20, 34 x 20 at the drawn marks. | B-24 x2 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 1559 | SK-1, SK-7; QC-07; B-24 | 0.80 / 5.8 |
| 7.18 | B-25: guy bridle on the crown top bar x1 | 412 mm (16.2 in). Folds: tab fold R (toggle loop); bar sleeve R start; bar sleeve R lap (bartack); bar sleeve L lap (bartack); bar sleeve L start; tab fold L (toggle loop). Box-X: 21 x 15, 21 x 15 at the drawn marks. Toggle tabs: fold each end on its 'tab fold' line and sew the drawn tip-end box (DECISION B25-BOX: the clear loop passes the POM toggle). FRAME COORDINATION: the bar sleeves close round the crown top bar BEFORE the crown corners are bonded (H1 build); the two sleeve-lap bartacks (BT-M) are sewn at frame assembly, not here. | B-25 x1 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 247 | SK-1, SK-7; QC-07; B-25 | 0.20 / 1.4 |
| 7.19 | TG: port guy (B-25 toggle -> B-27 port -> B-22 inside) x2 | 500 mm (19.7 in). Folds: -. Box-X: 21 x 25 at the drawn marks. | TG x2 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 315 | SK-1, SK-7; QC-07; TG | 0.20 / 1.4 |

| Seq | Operation | How (beginner) | Pieces | Machine | Needle | Thread | SPI | Stitch / seam | mm | Ref | h / SAM min |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 7.20 | K-G: guy tail keeper (25 mm elastic) x2 | Cut 90 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes). | K-G x2 | ZZ | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 304 / BT (zigzag lap) | | | 0.20 / 1.4 |
| 7.21 | GBL: guy bar loop (POM toggle) x2 | 170 mm (6.7 in). Folds: fold round the bar. Box-X: 21 x 25 at the drawn marks. | GBL x2 | BT / PS (prototype WF2 + J1) | Nm 140/22 DPx17, round point R | BN tex135 | 7 | 301 box-X path / BOX | 315 | SK-1, SK-7; QC-07; GBL | 0.20 / 1.4 |
| 7.22 | HPL: scope-pod hard-point loop x4 | 80 mm (3.1 in). Folds: fold, raw ends in the seam. Box-X / bartack per the load table where it is caught in a seam. | HPL x4 | BT | Nm 110/18 DPx17, round point R | BN tex90 | - | 304 / BT | | SK-1, SK-7 | 0.40 / 2.9 |
| 7.23 | E5: pole grip tab (cord lock) x1 | 60 mm (2.4 in). Folds: fold. Box-X / bartack per the load table where it is caught in a seam. | E5 x1 | BT | Nm 110/18 DPx17, round point R | BN tex90 | - | 304 / BT | | SK-1, SK-7 | 0.10 / 0.7 |
| 7.24 | I2T: IMP-2 hang tab (B-06f) x1 | 100 mm (3.9 in). Folds: fold. Box-X / bartack per the load table where it is caught in a seam. | I2T x1 | BT | Nm 110/18 DPx17, round point R | BN tex90 | - | 304 / BT | | SK-1, SK-7 | 0.10 / 0.7 |
| 7.25 | I2K: IMP-2 bottom keeper tab (B-01) x2 | 60 mm (2.4 in). Folds: fold. Box-X / bartack per the load table where it is caught in a seam. | I2K x2 | BT | Nm 110/18 DPx17, round point R | BN tex90 | - | 304 / BT | | SK-1, SK-7 | 0.20 / 1.4 |
| 7.26 | BRL: bow riser loop (through B-11) x2 | 100 mm (3.9 in). Folds: fold. Box-X / bartack per the load table where it is caught in a seam. | BRL x2 | BT | Nm 110/18 DPx17, round point R | BN tex90 | - | 304 / BT | | SK-1, SK-7 | 0.20 / 1.4 |
| 7.27 | BRS: bow riser strap + G-hook x1 | 300 mm (11.8 in). Folds: G-hook fold. Box-X / bartack per the load table where it is caught in a seam. | BRS x1 | BT | Nm 110/18 DPx17, round point R | BN tex90 | - | 304 / BT | | SK-1, SK-7 | 0.10 / 0.7 |
| 7.28 | MK-cord: rifle muzzle keeper, 3 mm shock cord loop through the 2 B-11 muzzle tabs + spring cord lock (one per gusset) x2 | Cut 309 mm (12.2 in), melt / whip both ends; fitted at hardware assembly (op below) as its use says. | MK-cord x2 | bench | - | - | - | - / - | | | 0.06 / 0.4 |
| 7.29 | F-EL: fly hem elastic, 6 mm (spine top + bottom, sides bottom + frame) x1 | Cut 2436 mm, lap 15 mm into a loop round its strap, zigzag the lap (3 passes). | F-EL x1 | ZZ | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 304 / BT (zigzag lap) | | | 0.10 / 0.7 |
| 7.30 | Fit the hardware | IM1/IM2: G-hook right end, cam threaded; IM3: wiregate carabiner on the free end; US1/US2: cam, then the STAINLESS 25 mm tri-glide threaded behind the cam as the cradle slip backstop (orchestrator 2026-11-13; gate US-BS); T3 loops: wiregate carabiner + TPU sleeve; W1-W3: rated G-hook end, heavy ladder lock on the gusset side; guys: toggle in the B-25 tab, through the B-27 port, heavy ladder lock on LLT, nylon tri-glide tail keeper; B-28: toggle rear, ladder lock both ends, 2 elastic keepers; RT: side-release buckles; E5: cord lock; cords MK-cord per their use. | hardware | bench | - | - | - | - / - | | QC-11 | 1.00 / 7.2 |
| 8.01 | Scope pod A1 | Box pleats in A1-b, leading-edge bar A1-d on the sleeve mouth (EFa), sleeve face to back panel A1-a round the perimeter; 4 hooks / loops to the HPL hard points. | A1-a, A1-b, A1-d | WF1 | Nm 110/18 DPx17, round point R | BN tex90 | 8 | 301 / SSa2 + EFa | 3770 | | 1.63 / 11.8 |
| 8.02 | Meat liner MS (SWAP slab) | Tube MS-a to frame face MS-b and floor MS-c; HDPE MS-s in the roll pocket. Silpoly: clips, no pins; 10 SPI tex 70. | MS-a, MS-b, MS-c, MS-s x2 | WF1 | Nm 90/14 DPx17, round point R | BN tex70 | 10 | 301 / SSa2 | 5280 | | 2.08 / 15.0 |
| 9.01 V | F-01 fly: panel seams | Join the fly panels (F-01a spine R \| F-01b side 1271 mm; F-01a spine L \| F-01b side 1271 mm). Silpoly: weights and clips, no pins; 301 then flat-fell; seam-seal the felled seams (piece note: silpoly: seams 301 then flat-felled + seam-sealed; 20 mm elastic channel hems top + bottom) | F-01a, F-01b | WF1 | Nm 90/14 DPx17, round point R | BN tex70 | 10 | 301 / SSa then LSc flat-fell (ISO 4916 class 2) | 7628 | | 3.05 / 22.0 |

| Seq | Operation | How (beginner) | Pieces | Machine | Needle | Thread | SPI | Stitch / seam | mm | Ref | h / SAM min |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 9.02 V | F-01 fly: elastic channel hems | Turn the hems 20 mm on the drawn hem lines (3249 mm of hem); thread 2436 mm of 6 mm elastic (F-EL), bar each end. | F-01a, F-01b, F-EL | WF1 | Nm 90/14 DPx17, round point R | BN tex70 | 10 | 301 / EFb channel | 3248 | | 2.35 / 16.9 |
| 10.01 | Trim, seal, turn | Trim thread ends to 3 mm, heat-seal webbing ends that frayed, turn the bag right side out through the door. | | bench | - | - | - | - / - | | | 0.50 / 3.6 |
| 10.02 | Final inspection and weigh | QC sheet sections F and M; weigh each kit against the ledger. | | bench | - | - | - | - / - | | QC-12 to QC-20 | 0.50 / 3.6 |
| 10.03 | Pack | README.md, Packaging section. | | bench | - | - | - | - / - | | | 0.20 / 1.4 |

# 7. Load points: bartack and box-X specification

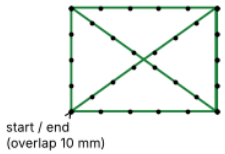
Box sizes are the box-X marks drawn on the pattern pieces. Load = factored design load from the record named (x 3.0 on the 1 g simulation per HA-LS-1 rev C). Capacity by attach_system rules. Proof = the test that releases the point (register).

| Point | Part | Where | Pattern | Stitches | Machine | Thread / needle | Load N (lbf) | Source | Capacity N | Governs | SF | Proof |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| port guy, toggle end | TG | TG toggle loop (B-25 tab side) | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 288.0 (65.0) | corrie_f_body.md 13.5: 96.1 N x 3.0 | 3186.0 | box-X | 11.06 | LL-H (366 N, toggle-loop pull, loadpath A3) |
| port guy, bag end | LLT + B-22 + B-02h | inside the hinge band at x +/-130.3 | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 288.0 (65.0) | corrie_f_body.md 13.5: 96.1 N x 3.0 | 1024.0 | B-22 perimeter 50 mm (primary body) | 3.56 | LL-H; PT-5 (lock + keeper); alternate body fabric: B-22 seam 1408 N |
| port guy, B-27 slit ends | B-01 | both ends of each slit, z 595 / 745 | bartack 10 x 2 mm, 304 | 28 | BT LK-1900BN | bonded_nylon_tex90 / Nm 110/18 DPx17, round point R | - (-) | no rated load (lap only); port mock-up: 100 N guy pull through the lap | | - | | port mock-up (13.9) |
| crown strap tab (B-28s x2) | B-28s | toggle loop | box-X 21 x 15 mm (0.83 x 0.59 in), path 124 mm | 39 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 288.0 (65.0) | corrie_f_body.md 15.2: 288 N envelope (port guy) if a crown strap ever holds the top | 2504.0 | box-X | 8.69 | LL-H (heavy lock slip 366 N covers 288 N; BL-031 dropped the 20 mm gate) |
| crown strap tab (B-28t x4) | B-28t | lock tab | box-X 21 x 20 mm (0.83 x 0.79 in), path 140 mm | 43 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 288.0 (65.0) | corrie_f_body.md 15.2: 288 N envelope (port guy) if a crown strap ever holds the top | 2836.0 | box-X | 9.85 | LL-H (heavy lock slip 366 N covers 288 N; BL-031 dropped the 20 mm gate) |
| crown strap front tab on the band | B-02h | B-28 front lock tab, OUTSIDE the hinge band (DECISION B28-FACE, BL-034), box through tab + band | box-X 21 x 20 mm (0.83 x 0.79 in), path 140 mm | 43 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 288.0 (65.0) | corrie_f_body.md 15.2 (as above) | 2836.0 | box-X | 9.85 | LL-H |
| B-25 bridle toggle tab | B-25 | toggle tab R: tip-end box, clear loop for the POM toggle (DECISION B25-BOX, BL-034) | box-X 21 x 15 mm (0.83 x 0.59 in), path 124 mm | 39 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 288.0 (65.0) | corrie_f_body.md 13.5 (guy) | 2504.0 | box-X | 8.69 | LL-H toggle-loop pull 366 N |
| B-25 bridle toggle tab | B-25 | toggle tab L: tip-end box, clear loop for the POM toggle (DECISION B25-BOX, BL-034) | box-X 21 x 15 mm (0.83 x 0.59 in), path 124 mm | 39 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 288.0 (65.0) | corrie_f_body.md 13.5 (guy) | 2504.0 | box-X | 8.69 | LL-H toggle-loop pull 366 N |
| B-25 bar sleeve laps | B-25 | bar sleeve R lap (bartack); bar sleeve L lap (bartack) (sewn at frame assembly, round the crown top bar) | bartack 16 x 2 mm, 304 | 42 | BT LK-1900BN | bonded_nylon_tex90 / Nm 110/18 DPx17, round point R | - (-) | no rated load: the sleeve wraps the bar, the lap only keeps it closed | | - | | LL-H toggle-loop pull (whole bridle) |
| guy bar loop | GBL | round the crown top bar | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 244.0 (55.0) | loadpath_verification A3: 244 N | 3186.0 | box-X | 13.06 | LL-H |
| IM3 bag-hook tongue | T3 | loop 34x30 | box-X 34 x 30 mm (1.34 x 1.18 in), path 219 mm | 65 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 1702.0 (383.0) | loadpath_verification A1: IM3 1702 N factored (guys-off misuse) | 4000.0 | carabiner (PT-1 basis: 0.5 x 8 kN minor axis) | 2.35 | PT-1 2553 N carabiner; PT-2 2553 N |
| IM3 bag-hook tongue | T3 | box-X to B-17 34x34 | box-X 34 x 34 mm (1.34 x 1.34 in), path 232 mm | 68 | PS AMS-210EN (or WF2 + J1) | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 1702.0 (383.0) | loadpath_verification A1: IM3 1702 N factored (guys-off misuse) | 4000.0 | carabiner (PT-1 basis: 0.5 x 8 kN minor axis) | 2.35 | PT-1 2553 N carabiner; PT-2 2553 N |

| Point | Part | Where | Pattern | Stitches | Machine | Thread / needle | Load N (lbf) | Source | Capacity N | Governs | SF | Proof |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B-17 IM3 tab perimeter (double row) | B-17 | OUTSIDE B-01 at x +/-115, z 385-455 | double-row perimeter 3 / 9 mm, 301 | | WF2 | bonded_nylon_tex90 / Nm 110/18 DPx17, round point R | 1702.0 (383.0) | loadpath_verification A1 | 3277.0 | perimeter 80 mm loaded, 2 rows | 1.93 | PT-2 2553 N on the 70 mm tab; perimeter 3 / 9 mm, 8 SPI tex 90; SF from the 90 x 110 basis, PT-2 settles the 70 mm tab |
| IM fairlead eyes (6) | IME | frame side, captive in the printed fairleads | box-X 34 x 30 mm (1.34 x 1.18 in), path 219 mm | 65 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 1702.0 (383.0) | IM3 eye carries the bag hook (A1 1702 N); IM1/IM2 605 N (S1) | 4430.0 | box-X | 2.6 | MT-1 analogue at 38 mm; IM3-EP 2553 N |
| hinge eye | HE + B-18 + B-01 | x +/-115, eye top z 60 | box-X 34 x 34 mm (1.34 x 1.34 in), path 232 mm | 68 | PS AMS-210EN (or WF2 + J1) | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 795.0 (179.0) | loadpath_verification A5: 795 N | 2355.0 | B-18 seam (80 mm, B-05 strip) | 2.96 | PT-3 1530 N |
| IM1 end loop | IM1 | left end loop, girth-hitched on the eye (box drawn at the tip end) | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 605.0 (136.0) | loadpath_verification S1: 605 N | 3186.0 | box-X | 5.27 | MT-1 908 N; MT-1 specimens are sewn exactly this way |
| IM2 end loop | IM2 | left end loop, girth-hitched on the eye (box drawn at the tip end) | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 605.0 (136.0) | loadpath_verification S1: 605 N | 3186.0 | box-X | 5.27 | MT-1 908 N; MT-1 specimens are sewn exactly this way |
| IM3 end loop | IM3 | left end loop, girth-hitched on the eye (box drawn at the tip end) | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 605.0 (136.0) | loadpath_verification S1: 605 N | 3186.0 | box-X | 5.27 | MT-1 908 N; MT-1 specimens are sewn exactly this way |
| US1/US2 sewn eye | US | eye on the left pin (US-EYE fold, pattern) | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 720.0 (162.0) | loadpath_verification S5-S7: 720 N | 3186.0 | box-X | 4.43 | MT-1 1080 N; ; cam side: stainless tri-glide backstop behind the cam (orchestrator 2026-11-13; gate US-BS) |
| W1 G-hook end | W1 | G-hook fold 2 x 30 mm | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 225.0 (51.0) | loadpath_verification S9: 225 N | 3186.0 | box-X | 14.16 | LL-H 366 N; W-PULL 338 N |
| W2 G-hook end | W2 | G-hook fold 2 x 30 mm | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 225.0 (51.0) | loadpath_verification S9: 225 N | 3186.0 | box-X | 14.16 | LL-H 366 N; W-PULL 338 N |
| W3 G-hook end | W3 | G-hook fold 2 x 30 mm | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 225.0 (51.0) | loadpath_verification S9: 225 N | 3186.0 | box-X | 14.16 | LL-H 366 N; W-PULL 338 N |
| W haul extender | WX | box-X hook end | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 225.0 (51.0) | ASSUMPTION WX-LOAD: same float-clamp load as S9 (225 N) | 3186.0 | box-X | 14.16 | LL-H; one extender pulled to 338 N |
| W haul extender | WX | box-X loop | box-X 21 x 25 mm (0.83 x 0.98 in), path 157 mm | 48 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 225.0 (51.0) | ASSUMPTION WX-LOAD: same float-clamp load as S9 (225 N) | 3186.0 | box-X | 14.16 | LL-H; one extender pulled to 338 N |
| B-24 float link: hook end | B-24 | hook-end fold (boxes as drawn by attach_system.link_piece) | box-X 34 x 30 mm (1.34 x 1.18 in), path 219 mm | 65 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 941.0 (212.0) | loadpath_verification S12 / RD-02: 941 N misuse | 4000.0 | carabiner | 4.25 | PT-6 1410 N |
| B-24 float link: loop bases | B-24 | 3 loop bases (boxes as drawn by attach_system.link_piece) | box-X 34 x 20 mm (1.34 x 0.79 in), path 187 mm | 56 | BT LK-1900BN | bonded_nylon_tex135 / Nm 140/22 DPx17, round point R | 941.0 (212.0) | loadpath_verification S12 / RD-02: 941 N misuse | 3786.0 | box-X | 4.02 | PT-6 1410 N; loop boxes close each T3 loop across both plies |

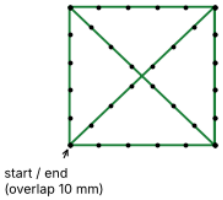
## SK-1  Load-point stitch patterns (true proportions; dots = 7 SPI stitch positions)

box-X 21 × 15 mm (0.83 × 0.59 in)



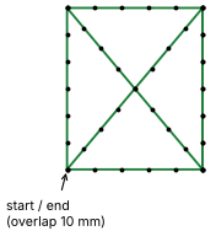

39 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: B-25, B-28s

box-X 21 × 20 mm (0.83 × 0.79 in)



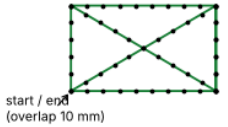

43 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: B-02h, B-28t

box-X 21 × 25 mm (0.83 × 0.98 in)



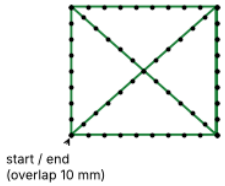

48 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: GBL, IM1, IM2, IM3, LLT + B-22 + B-02h, TG, US, W1, W2, W3, WX

box-X 34 × 20 mm (1.34 × 0.79 in)



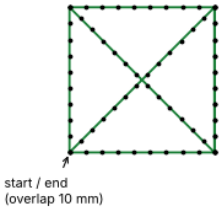

56 stitches incl. 4 lock, 7 SPI, tex 135, Nm 14
used on: B-24

box-X 34 × 30 mm (1.34 × 1.18 in)




65 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: B-24, IME, T3

box-X 34 × 34 mm (1.34 × 1.34 in)




68 stitches incl. 4 lock, 7 SPI, tex 135, Nm 140 R
used on: HE + B-18 + B-01, T3

BT-S bartack 10 × 2 mm, 28 st (304)



LK-1900 standard tack, tex 90, Nm 110 R
B-27 slit ends, FP mouth corners

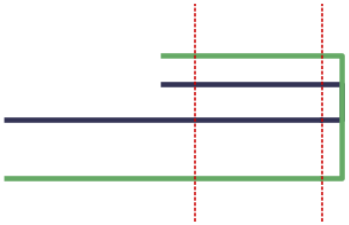
BT-M bartack 16 × 2 mm, 42 st (30



LK-1900 standard tack, tex 90, Nm 110 R
I2K, HE neck, lid channel ends,
B-25 sleeve laps

## SK-2 Seam constructions (cross-sections, not to scale; red dashed = stitch row)

SSa2 body seam: 12 mm, then allowances turned and caught at 6 mm (2 rows)



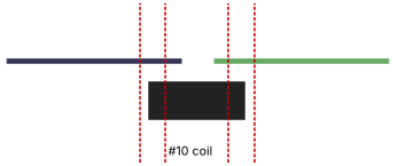
LSd Hypalon patch: double row 3 / 9 mm from the edge, 8 SPI tex 90



EFb roll-top hem round the HDPE strip



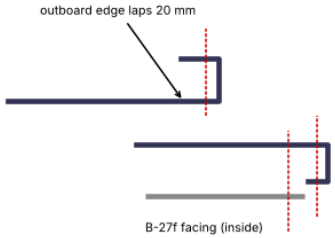

ZIP: tape sewn at 5 mm, turned, edge-stitched 3 mm (door, lid, pockets)



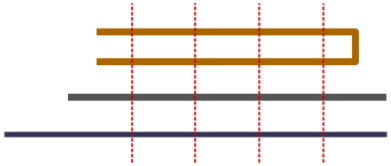

SSa-ins: loop / tab caught in the seam, backtack each side (HPL, K-W, E5)



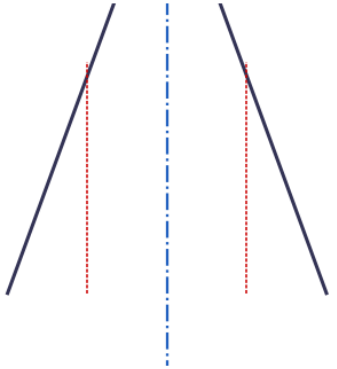

B-27 guy port: both edges turned 6 mm, facing on the inboard edge, bartack ends



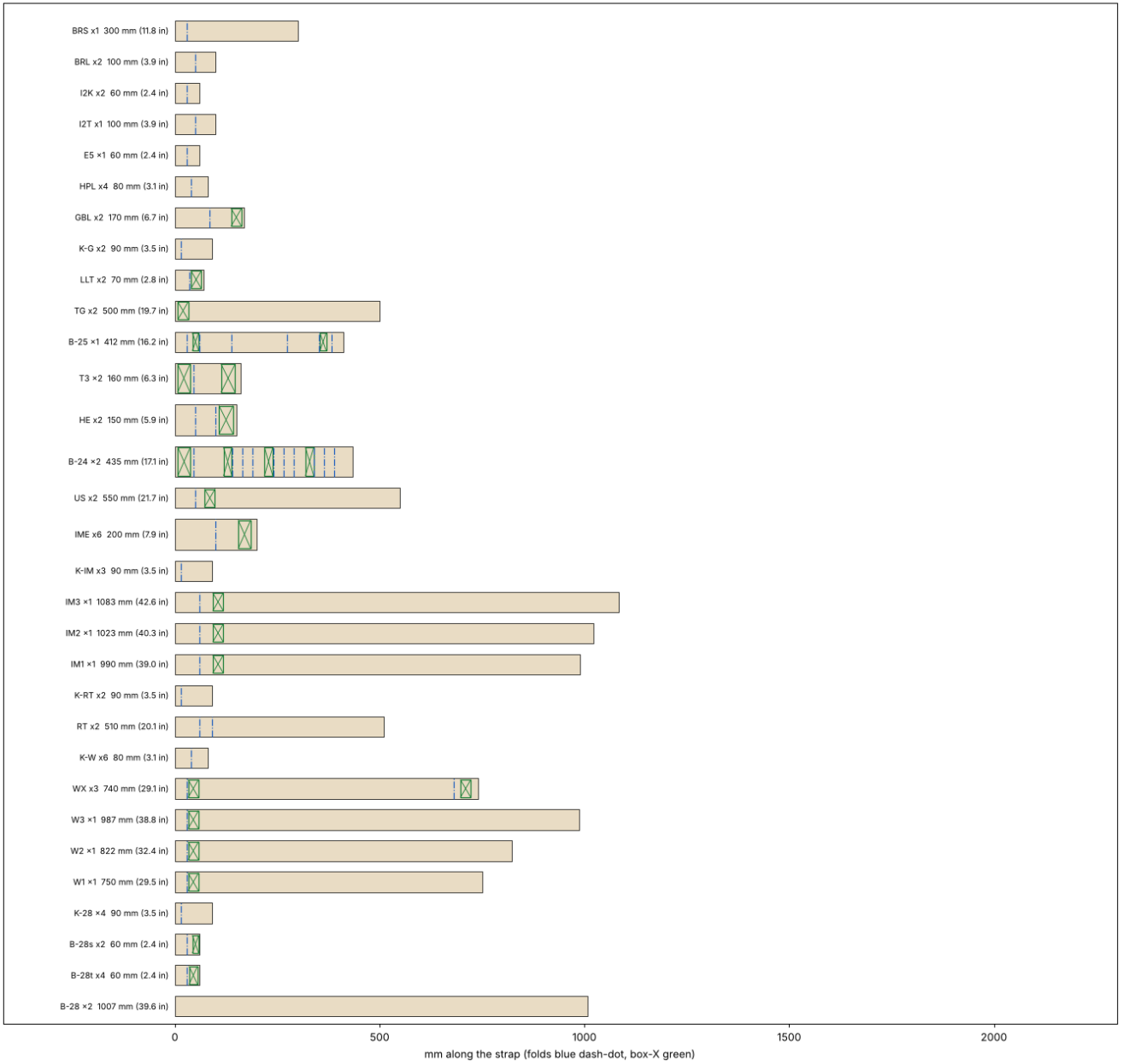

Load stack (e.g. LLT + B-22 + B-02h): webbing fold / Hypalon / panel, box-X through all



DART (B-29a, 4 corners): fold on the centre, sew leg to leg, tie off at the apex



SK-7  Webbing parts as cut (crown kit M); hot-knife length = strap length



# 8. QC checkpoints, release tests, defects

| ID | Stage | Check | Method | Pass | Frequency |
|---|---|---|---|---|---|
| QC-01 | Print | Test square | ruler on 00_PRINT_FIRST and on one page per print run | 100.0 +/- 1.0 mm both ways | every print run |
| QC-02 | Incoming | Fabric weight and width | cut 100 x 100 mm, weigh 0.01 g; tape the roll width | gsm within +/- 5 % of the record; width >= usable_width_mm | every roll |
| QC-03 | Cutting | Piece size and cut-outs | lay the pattern on the cut piece | +/- 1 mm on pieces < 300 mm, +/- 2 mm on larger; windows / slots / slits present and clean | first piece of each lay + 1 in 10 |
| QC-04 | Cutting | Notches, grain, webbing length | visual; tape | notch 6 mm (never past the seam line); grain within 2 deg of the arrow; webbing +/- 2 mm, ends sealed | first of each + 1 in 10 |
| QC-05 | Sewing | Stitch density | count stitches over 25.4 mm | body 8, light 10, box-X 7 SPI; +/- 1 | start of each shift / each op set-up |
| QC-06 | Sewing | Tension balance | cut a swatch seam open | lock mid-stack, no loops on either face | each set-up |
| QC-07 | Load points | Box-X and bartacks | visual against SK-1; count | full rectangle + both diagonals, start / end overlap >= 10 mm, stitch count >= load table, no skipped stitch, needle hole not torn | 100 % |
| QC-08 | Assembly | Seam match | notches at both ends of every 50 mm | notch offset <= 2 mm; seam end mismatch <= 3 mm; allowance 12 +/- 1 mm | 100 % |
| QC-09 | Assembly | Door zip | run both sliders 10 x end to end; look at the corners | no snag, no wave > 3 mm at the R 55 corners, tape not caught, stops at the hinge band | 100 % |
| QC-10 | Assembly | B-27 ports | pull the guy through at 45 deg; look inside | 20 mm lap closes flat; both ends bartacked; facing follows the crease at z 631 | 100 % |
| QC-11 | Hardware | Hardware fitted | count against the BOM; threading as SK-7 | every lock threaded the right way (pull test by hand), carabiners gate-in, cord lock holds | 100 % |
| QC-12 | Final | Width | tape across the widest point, bag packed full | <= 394 mm (15.5 in) (owner limit) | first article + 1 in 10 |
| QC-13 | Final | Volume | fill with 10 mm foam balls / litre jug method, collar rolled 3 times | 85 L +/- 3 L (5200 cu in) (owner limit) | first article |
| QC-14 | Final | Mass, bag base | 1 g scale | within +/- 3 % of the ledger base bag (1133 g on the ledger basis; primary fabrics are heavier, see costing.md) | every bag |
| QC-15 | Final | No hook-and-loop, coil zips only | visual | zero hook-and-loop anywhere (owner: quiet bag) | 100 % |
| QC-16 | Final | Camo use | visual | camo only on B-02u, SH-07, SH-01b, FP-b, B-29f, A1-d (accents) | 100 % |
| QC-17 | Final | Thread ends, seals | visual | ends <= 3 mm, no open webbing ends, no burnt fabric | 100 % |
| QC-18 | Final | Labels | visual | ID label with rev 3 + date, size / crown kit, load-limit card (53 kg frame meat, RD-02) | 100 % |
| QC-19 | Final | Kits complete | count | base / as-hunted (lid, B-28 per kit, scope pod) / meat-trip (WX, B-24, liner) bags each complete | 100 % |
| QC-20 | Final | Fit on the frame | hook to a frame or the fit jig: hinge pins, IM eyes through B-12 ports, guys through B-27 | every eye reaches its port without strain; door opens fully | first article + 1 in 25 |

# Physical tests that gate release

| ID | Test | Gates | Steps | Pass / fail | Source |
|---|---|---|---|---|---|
| MT-1 | 25 mm box-X bartack proof at the BL-010 per-row loads | every 25 mm box-X on IM1-IM3 and US1/US2 | sew 10 loops exactly like the pack; loop over the 5 mm pin; 5 loops to 908 N (92.6 kg) and 5 to 1080 N (110.1 kg); hold 1 min each | 5 of 5 hold 908 N (IM) and 5 of 5 hold 1080 N (US) for 1 min, no broken stitch | test_register.csv |
| MT-2 | meat-strap hook proof per strap (BL-010) | IM free-end hooks | as PT-1: G-hooks to 645 N (65.7 kg, covers IM1 410 N), carabiners to 910 N (92.8 kg); hold 1 min | 5 of 5 G-hooks hold 645 N and 5 of 5 carabiners hold 910 N for 1 min, gates close | test_register.csv |
| MT-3 | 25 mm cam buckle slip at the BL-010 per-row loads | IM / US cams | thread the cam as in the pack; paint-pen mark at the cam mouth; pull to 908 N (92.6 kg), hold 1 min, measure; then to 1080 N (110.1 kg), hold 1 min, measure | mark moves < 3 mm at 908 N and at 1080 N, 5 of 5 | test_register.csv |
| PT-1 | IM3 bag carabiner on the sewn T3 tongue, per-row proof (BL-010) | IM3 carabiner on T3 | carabiner over the 5 mm steel pin; sewn T3 tongue on the carabiner; pull with the come-along to 2553 N (260.2 kg on the scale); hold 60 s; release; check the gate closes and look for set | 5 of 5 hold 2553 N for 60 s, gate closes after, no visible set | test_register.csv |
| PT-2 | T3 tongue on the 70 mm B-17 tab on a B-01 swatch (BL-010 load) | T3 on the 70 mm B-17 tab | clamp the B-01 swatch 120 mm wide between bar clamps; tongue loop on the pin; pull to 2553 N (260.2 kg); hold 60 s | 5 of 5 hold 2553 N, no stitch break, no laminate tear | test_register.csv |
| PT-3 | HE eye on B-18 + B-01 + B-05 seam, built without binding | HE eye on B-18 (no binding) | 5 mm pin through the eye; clamp the swatch; pull to 1530 N; hold 60 s | 5 of 5 hold 1530 N | test_register.csv |
| PT-5 | guy ladder lock + tri-glide keeper slip | guy lock + tri-glide keeper | mark webbing; 10 cycles 0 -> 170 N -> 0 by hand on the come-along; measure the mark | slip < 5 mm over 10 cycles | test_register.csv |
| PT-6 | B-24 float stand-off link | B-24 float link | link hook on the pin; 38 mm loop through the 220 mm loop; pull in line to 1410 N; hold 60 s | 5 of 5 hold 1410 N, no box-X break, gate closes | test_register.csv |
| LL-H | heavy 25 mm ladder lock slip, 1 h hold (guys and W1-W3) | heavy 25 mm ladder lock slip: guys, W1-W3 and the B-28 crown straps (the B-28 slip gate since BL-031) | thread as in the pack; paint-pen mark at the lock; hang 37.3 kg (366 N, sand bags) 50 mm above the floor for 1 h, all 5 side by side; measure the mark | slip < 5 mm after 1 h, 5 of 5 (366 N covers the W 338 N proof) | test_register.csv |
| T9 | loaded CG (+ 15 in, 30 kg meat check); CORRIE F rev 3 items in rows T9-R3a..d (BL-038 split) | loaded CG on the first article (rev C basis); its rev 3 items are the T9-R3a..d rows below (BL-038 split) | 68 kg fill and meat on IM1-IM3 at 15/19/23 in; hang from haul loop, then from a hip pin, plumb both lines; add 30 kg on IM1-IM3 at 15 in; then rows T9-R3a (port guys, S kit CG), T9-R3c (crown strap path) in the same session; T9-R3b (cinch after a walk) and T9-R3d (lid) after T1-T4 | bag CG <= 205 mm at z <= 590; meat <= 190; tall <= 185; rev 3 items: rows T9-R3a..d | test_register.csv |
| T9-R3a | T9 rev 3 (a): port guys hold the S-kit top, meat CG at 15 in | rev 3 (a): B-27 port guys hold the S-kit top, CG at the S kit (BL-030) | top packed full above the guys; 30 kg sand meat on IM1-IM3, fill to 68 kg; hang from the haul loop and plumb a line; hang from a pin through the hip pivot and plumb a second line; the CG is where the lines cross; measure its distance aft of the stay plane | meat CG <= 190 mm aft (record against 179.3 mm) | test_register.csv |
| T9-R3b | T9 rev 3 (b): crown strap cinch force and retention (CROWN_PRE_N, MU_LID) | rev 3 (b): B-28 crown strap cinch force and retention | hook the spring scale on each B-28 tail; pull to 8.2 kg (80 N) and lock; walk 30 min loaded (only after T1-T4 pass); read the tension again (pull the tail until it just moves); set the pack down hard from 0.5 m on its base; read again; check the lid is still seated | after the walk >= 40 N, or the lid still seated on the full top (report both values) | test_register.csv |
| T9-R3c | T9 rev 3 (c): crown strap free length (ROLL-PATH, TORSO-LIN) | rev 3 (c): B-28 crown strap free length per crown kit (cut = longest path + 40 mm spare + 2 grips, BL-039) | cinch both crown straps; measure the strap path from the bar tab to the ladder lock along the strap with a tape (both straps); repeat at the L kit 6-day setting | S kit full: path <= 968.1 mm (cut 1068.1 - 2 x 50 mm grip tail; sim 928.1 mm); L kit 6-day: record against the sim's 438.8 mm | test_register.csv |
| T9-R3d | T9 rev 3 (d): sewn lid mass and seat (LID_SLIDER5_G, LID_THREAD, LID-SEAT) | rev 3 (d): sewn B-29 lid mass and seat | weigh the lid alone 3 times; photograph it seated on the packed 6-day bag and the full bag, from the rear 3/4 and the side | lid 47.8 g +/- 5 g (ledger); lid edge on the hinge-band seam, camo caps over the blades | test_register.csv |
| FS-O5 | ULTRA 200X 50 mm strip tensile (C-5 / O-5) | alternate fabric gate: ULTRA 200X 50 mm strip (floor / frame side) | clamp each strip in the 50 mm bar grips, 200 mm gauge; pull slowly to break; record the peak | mean >= 1840 N/50 mm, minimum >= 1195 N/50 mm | test_register.csv |
| B-28 slip | B-28 crown strap on its heavy ladder lock: covered by LL-H at 366 N (BL-031 dropped the 20 mm gate; 25 mm strap on ladder_lock_25mm_heavy) | release of this package | B-28 crown strap on its heavy ladder lock: covered by LL-H at 366 N (BL-031 dropped the 20 mm gate; 25 mm strap on ladder_lock_25mm_heavy) | LL-H pass | corrie_f_body.md 15.6 / 16 |
| PORT-1 | B-27 port mock-up: hose from above 1 min with the guy pulled at 45 deg; 100 N guy pull through the lap; gloved lock use at the port | release of this package | B-27 port mock-up: hose from above 1 min with the guy pulled at 45 deg; 100 N guy pull through the lap; gloved lock use at the port | no water inside; lap stays closed at 100 N | corrie_f_body.md 13.9 |

| ID | Test | Gates | Steps | Pass / fail | Source |
|---|---|---|---|---|---|
| GT-1 | capstan pull: 5 kg on the guy over the packed bag's corner | release of this package | capstan pull: 5 kg on the guy over the packed bag's corner | record MU_WEB_FAB (0.20-0.40 band) | corrie_f_body.md 13.9 |
| ALT-1 | alternate set: owner sees an ULTRAGrid swatch | release of this package | alternate set: owner sees an ULTRAGrid swatch | owner OK in writing | part A README section 1 |
| ALT-2 | alternate set: PT-3 on 200X + 50 mm strip test | release of this package | alternate set: PT-3 on 200X + 50 mm strip test | >= 36.8 N/mm (1840 N / 50 mm mean, FS-O5) | part A README; register FS-O5 |
| ALT-3 | alternate set: weigh the 70D stock | release of this package | alternate set: weigh the 70D stock | gsm within +/- 5 % of nylon_ripstop_70d record | part A README |
| US-BS | US backstop: cam on 25 mm webbing with the stainless tri-glide threaded behind it; open the cam (forced slip) and load the strap to the US proof 1080 N (MT-1 basis, S5-S7 720 N x 1.5) | release of this package | US backstop: cam on 25 mm webbing with the stainless tri-glide threaded behind it; open the cam (forced slip) and load the strap to the US proof 1080 N (MT-1 basis, S5-S7 720 N x 1.5) | tri-glide stops the strap within 10 mm and holds 1080 N for 1 min, 3 of 3 (ASSUMPTION US-BS) | triglide_25mm record note; DECISIONS 2026-11-13 |
| FRAY-1 | wash-swatch fray test of the un-bound SSa2 seam (3 wash cycles, R-12) | release of this package | wash-swatch fray test of the un-bound SSa2 seam (3 wash cycles, R-12) | no allowance frayed past the 2nd row; else bind (ASSUMPTION NO-BIND) | RD-17 |

# Defect list

| Defect | Where | Cause | Prevent / fix | Class |
|---|---|---|---|---|
| Skipped stitches in a box-X | load points | needle too small / wrong point, flagging on Hypalon | Nm 140 R, tex 135, walking foot pressure; re-sew the whole box offset 1 mm | critical |
| Box-X short (stitch count under the table) | load points | early stop, wrong program | count against the load table; re-sew | critical |
| Hardware threaded backwards (ladder lock, cam) | straps | assembly | QC-11 hand pull; SK-7 | critical |
| Load tab on the wrong face (B-17 out, B-18 / B-22 in) | B-01, B-02h | face mix-up | placement labels say INSIDE / OUTSIDE; check before sewing | critical |
| Needle-cut Hypalon / laminate (perforation line) | B-17, B-18, B-22, B-01 | cutting point needle or SPI too high | R point only; 8 SPI max on Hypalon | critical |
| Swell puckered or twisted in its window | B-01 | window not notch-matched | unpick, re-clip at the corners | major |
| B-27 port gapes (lap < 20 mm) | B-01 | facing sewn on the wrong line | re-sew on the drawn stitch line | major |
| Door zip waves / snakes at the corners | door | zip stretched while sewing | basting tape, corner notches, ease into the corner | major |
| Seam allowance frayed past the 2nd row | body seams | straight-knife cut nylon | laser / hot-knife cut or bind (FRAY-1) | major |
| Notch mismatch > 2 mm / seam end mismatch > 3 mm | box seams, collar ring | pieces stretched | re-clip notch to notch, walking foot | major |
| Mesh pocket over-stretched | FP | mesh pulled while sewing | baste the pleats first, sew with the mesh on top | minor |
| Thread tension loops on the underside | any | tension / bobbin | QC-06 swatch at each set-up | minor |
| Burn marks / hard beads on webbing ends | webbing | hot knife too hot / slow | J8 stop, steady cut | minor |
| Camo piece cut against the print direction | B-02u, SH-* | lay error | ONE WAY lay rule | minor |
| Hook-and-loop or a non-coil zip fitted | any | substitution | BOM check (owner: quiet bag) | critical |

# 9. Packaging

1. Bag: collar unrolled, door zipped, straps coiled inside the bag (keepers on), folded flat to about 450 x 350 mm.
2. Base kit inside the bag: IM1-IM3, US1/US2, W1-W3 fitted; guys fitted to B-22 with the toggle end coiled.
3. As-hunted kit bag (poly bag 1, labelled): B-29 lid with the B-28 straps for the ordered crown kit (S / M / L), scope pod.
4. Meat-trip kit bag (poly bag 2, labelled 'MEAT TRIP KIT'): 3 WX extenders with G-hooks, 2 B-24 links with carabiners, meat liner MS.
5. Card on top: load limits (53 kg / 116 lb frame meat in float mode, RD-02), guy rule (port guys fitted for every carry, RD-23), the strap tail rule, wash cold / line dry, no hook-and-loop.
6. Outer: recycled poly bag 600 x 800 mm, 10 bags per carton; carton label: rev 3, door D2, crown kit, fabric set (primary / alternate), date, lot.

# 10. Costing and hours

| Step | Beginner h |
|---|---|
| 0 Prep and cutting | 16.19 |
| 1 Pockets | 6.37 |
| 2 Small parts | 3.97 |
| 3 Door D2 and hinge band | 5.00 |
| 4 Main bag | 13.54 |
| 5 Collar and roll top | 3.40 |
| 6 Crown lid B-29 (as hunted) | 8.46 |
| 7 Straps and webbing | 7.36 |
| 8 Add-ons (scope pod as hunted, meat liner meat trip) | 3.71 |
| 9 Rain fly F-01 | 5.40 |
| 10 Finish, inspect, pack | 1.20 |
| Total (incl. one-off set-up) | 74.6 |

| Scenario | SAM min | Labour $ | CMT incl. overhead $ | Materials $ | Total $ |
|---|---|---|---|---|---|
| us_contract_usd_min | 508 | 431.36 | 539.20 | 367.41 | 906.61 |
| asia_contract_usd_min | 508 | 60.90 | 76.12 | 367.41 | 443.53 |

Lid on the drafted pattern 9.53 h (booked 7.76 h). Alternate set materials $367.25. Assumptions: SAM-RATIO 0.12; LABOUR {'us_contract_usd_min': 0.85, 'asia_contract_usd_min': 0.12}; OVERHEAD 0.25; THREAD-FACTOR 2.75; WEB_WASTE 5 %; prices with no record: PRICE_ASSUMED.

# 11. Release delta register: rows this release resolves

| RD | Status | Evidence (checked by the script) | Still open |
|---|---|---|---|
| RD-01 | closed | F-03 in the rev 3 piece list: False | nothing (v3.1 set stays superseded with its STOP notice) |
| RD-02 | restated | B-24 x2 + 2 carabiners in the meat-trip kit BOM: True; PT-6 in the release gates | user guide (RD-22) with the 53 kg limit and belly rule; PT-6 |
| RD-03 | closed | IM1 cut 990 mm in the rev 3 cut list | MT-1 / MT-2 on the sewn straps |
| RD-12 | restated | B-17, B-18, B-22, HE, T3, LLT, TG drafted and sequenced: True; B-19 / B-20 / B-23 / T2 removed (BL-004 REC); IM3 hooks are wiregate carabiners (BL-010) | PT-1, PT-2, PT-3, PT-5, PT-6, LL-H before any load point is sewn on a bag that will carry |
| RD-15 | closed (numbers) | BOM, yardage and mass use the current records (robic 125 gsm, ultragrid 132 gsm 1500 mm roll, ULTRA 133 gsm) | physical: swatch weights (QC-02), FS-O5 / PT-3 on 200X (alternate gate) |
| RD-16 | closed (release) | CORRIE F rev 3 patterns (part A) + this op sequence replace SPINE step 9 | first article build; CORRIE gates; pattern-read ops listed as volatile ops |
| RD-17 | restated | IMP-2 / IMP-4 / IMP-8 in the bag: True; SH-03 absent: True; body seams un-bound (SSa2, ASSUMPTION NO-BIND) | FRAY-1 wash-swatch test; IMP-1/3/5/6 belong to the harness / belt packages |
| RD-18 | closed | B-11 x8 + BRL x2 + BRS + G-hook in the BOM: True; B-11 placements drawn on B-03: 8 tabs (op 4.07 released) | archery / rifle loadout fit settles ASSUMPTION B11-PLACE |
| RD-21 | closed (pointer) | this op sequence supersedes BUILD_PLAN.md step 9 for the bag; BUILD_PLAN.md points at the rev 3 release: True | BUILD_PLAN.md body still describes the SPINE bag (its banner says so) |
| RD-22 | open | the packaging card (README Packaging) carries the load limits | the user guide itself |
| RD-23 | restated | B-25, B-27f, B-22 inside, TG, LLT drafted and sequenced (ops 3.04-3.05, 4.03): True | T9 rev 3 at the S kit; LL-H; PORT-1; attorney review; B-22 placed at x +/-130.3 vs drawn 149 (ASSUMPTION B22-IN) |
| RD-24 | restated | B-29 (7 piece types) + B-28 per kit + keepers sequenced; lid 9.53 h on the drafted pattern; build_metrics books the op-sequence hours (build_effort_rev3.json: 70.49 h a bag) | T9 lid weigh / photo, cinch, free lengths |

# 12. Assumptions that govern

| ID | Assumption | Settled by |
|---|---|---|
| NO-BIND | Body seams are SSa2 (second row over the turned allowances), not bound: the ledger books no binding (RD-17 B-1). | FRAY-1 wash-swatch test (3 cycles); if it fails, bind with 20 mm grosgrain via a binder |
| NO-TAPE | No taped seams: the bag is water resistant, contents in the user's dry bag. | owner decision if a waterproof bag is wanted (seam-tape machine + taped ULTRA seams) |
| SAM-RATIO | Factory SAM = 0.12 x beginner hours (bag_concepts.HOURS basis). | time study on the first 10-bag lot |
| LABOUR | Loaded CMT minute: US contract $0.85/min, Asia $0.12/min; overhead + margin 25%. | two contract quotes from this tech pack |
| THREAD-FACTOR | 2.75 m of thread per m of 301 stitching (both threads, ends). | weigh the cones before / after the first bag |
| SPI-LIGHT | 10 SPI on 40D-100D and mesh. | QC-05/06 swatch: no puckering, no perforation |
| THREAD-NEEDLE | Tex 70 / Nm 90, tex 90 / Nm 110, tex 135 / Nm 140, round point (attach_system.NEEDLE for 90 / 135). | swatches in op 0.02: no skipped stitches, no shredding |
| UTIL-SMALL | Small and sheet pieces nest at 70% across bags in production. | first production markers |
| UTIL-ALT | Alternate-set body fabrics nest at 85% in production (no 10-bag alternate marker). | alternate 10-bag markers if the alternate set passes its gates |
| B11-PLACE / KW-PATH / F02-CENTRE / US-EYE | Pattern placements read by this package (patterns_corrie_f_rev3, BL-034 part A): B-11 tab heights, K-W keeper heights from the W path, F-02 fore-aft on B-05, the US eye fold. | archery / rifle loadout fits; T9 W free lengths (keeper within +/-15 mm of the wrap); fly fit; MT-1 specimens |
| WX-LOAD | Haul extenders carry the S9 float-clamp load (225 N). | pack-designer load case; LL-H |
| US-BS | The stainless 25 mm tri-glide behind each US cam stops cradle slip (record: 3.75 kN break, no slip rating). | gate US-BS: cam slipped on purpose, backstop holds |
| TG-SS-PRICE | Stainless tri-glide USD 2.84 from the record's EUR source line, converted as the record does (ex VAT x 1.09 USD/EUR). | supplier quote |
| BT-STD | Slit-end bartack 10 x 2 mm, 28 stitches; standard tack 16 x 2 mm, 42 stitches (LK-1900 patterns). | QC-07 |
| PRICES | Items with no material record use PRICE_ASSUMED (TPU hook sleeve, rubber pull sleeve, basting tape, seam sealer, labels, packaging). Hardware and elastic use their records since BL-034. | supplier quotes |